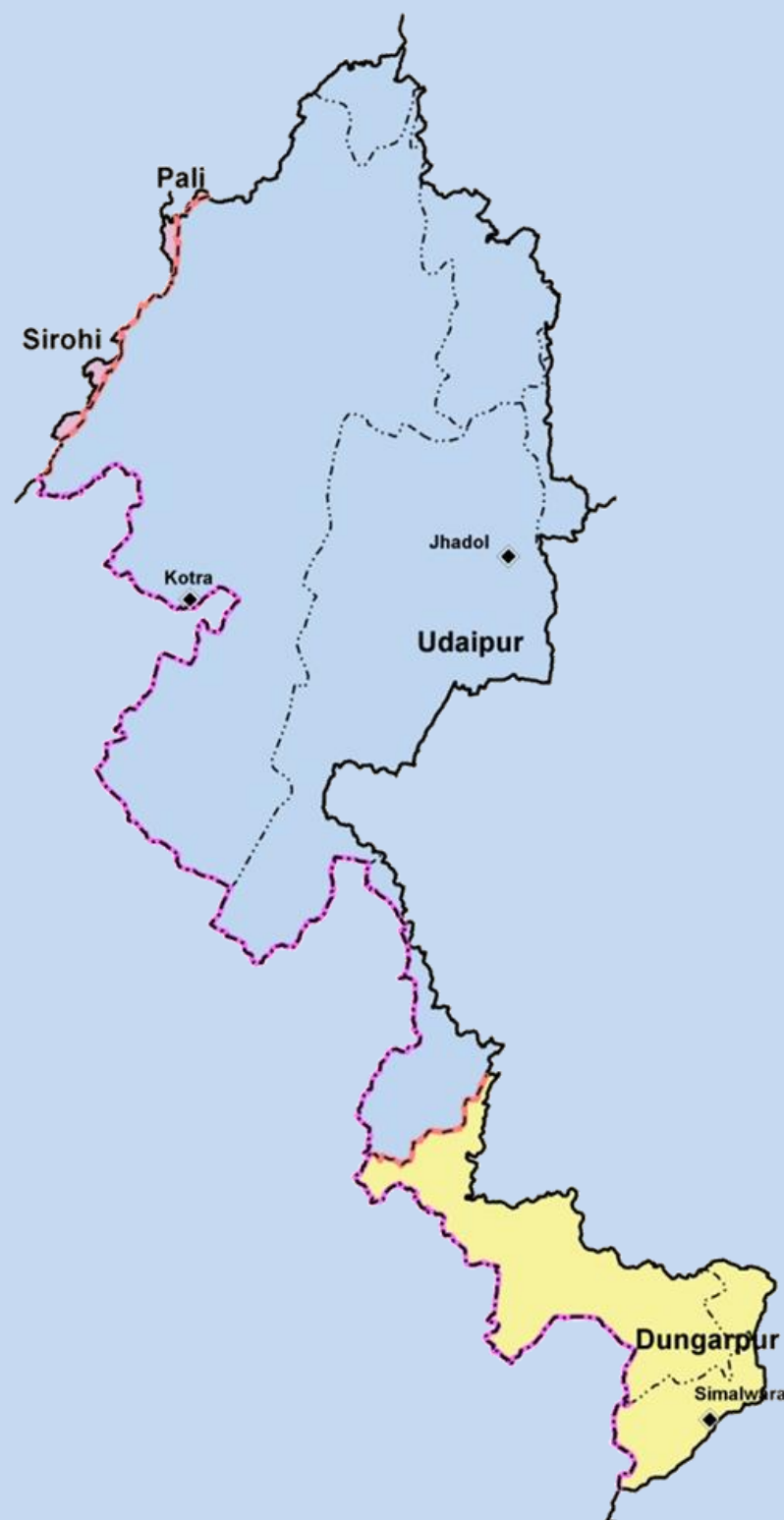


Hydrogeological Atlas of Rajasthan

Sabarmati River Basin



2013

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

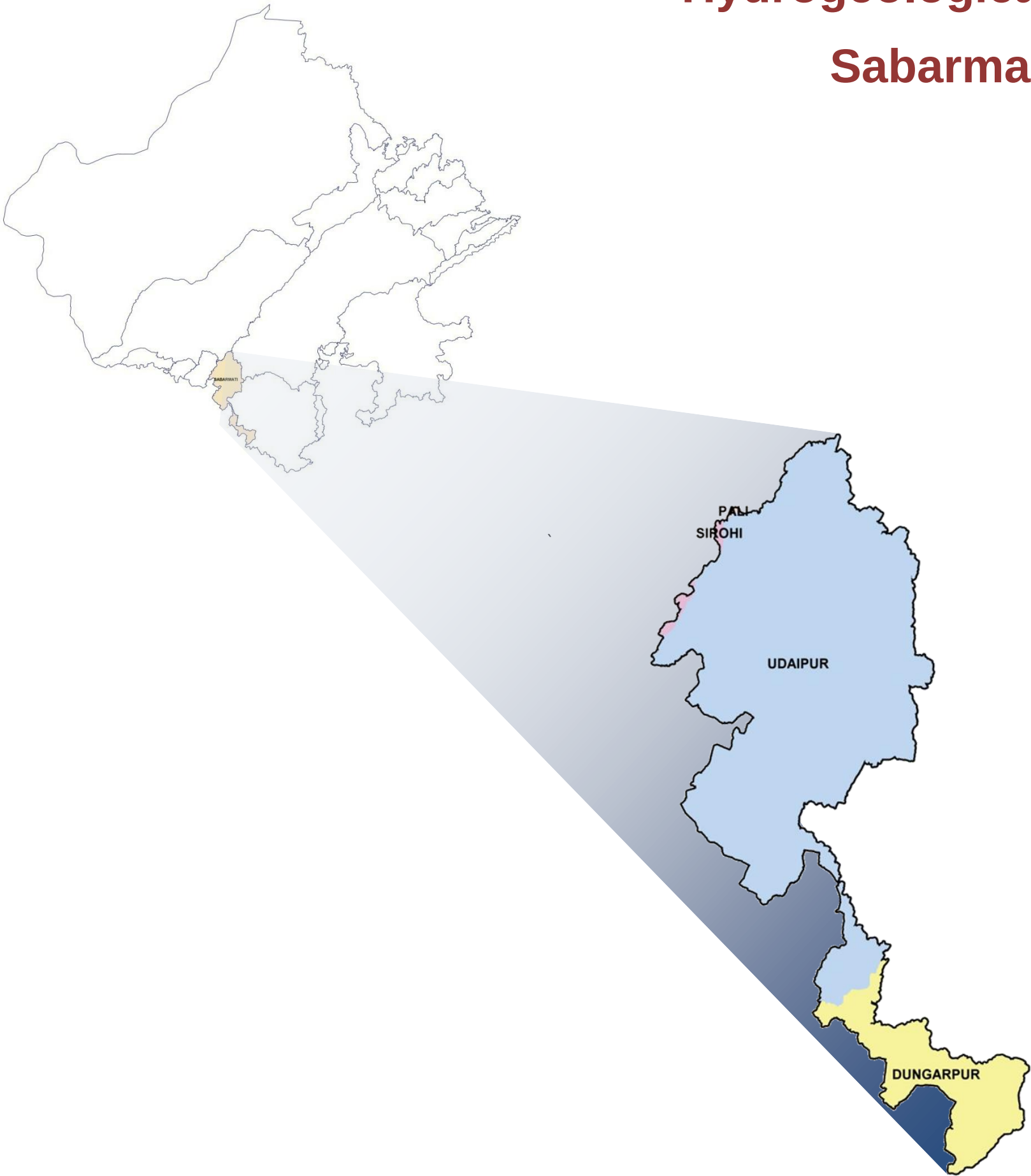
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Sabarmati River Basin



Contents:

List of Plates	Title	Page No.
Plate I	Administrative Map	2
Plate II	Topography	4
Plate III	Rainfall Distribution	4
Plate IV	Geological Map	6
Plate V	Geomorphological Map	6
Plate VI	Aquifer Map	8
Plate VII	Location of Ground Water Monitoring Stations	8
Plate VIII	Location of Exploratory Wells	10
Plate IX	Depth to Water Level (Pre-Monsoon 2010)	10
Plate X	Water Table Elevation (Pre-Monsoon 2010)	12
Plate XI	Water Level Fluctuation (Pre-Post Monsoon 2010)	12
Plate XII	Electrical Conductivity Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIII	Chloride Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIV	Fluoride Distribution (Average Pre-Monsoon 2005-09)	16
Plate XV	Nitrate Distribution (Average Pre-Monsoon 2005-09)	16
Plate XVI	Depth to Bedrock	18
Plate XVII	Map of Unconfined Aquifer	18
Plate XVIII	Index Map Showing Alignment of Cross Sections	20
Plate XIX	Cross Section Along A-A'	22
Plate XX	Cross Section Along B-B'	22
Plate XXI	Cross Section Along C-C'	24
Plate XXII	Cross Section Along D-D'	24
Plate XXIII	3D Model	26
Glossary of terms		27

ADMINISTRATIVE SETUP

SABARMATI RIVER BASIN

Location:

Sabarmati River Basin is located in mid south-western Rajasthan. It stretches between 23° 28' 25.18" to 24° 55' 14.72" North latitude and 72° 59' 38.90" to 73° 48' 29.62" East longitudes. In the northwest the boundary of Sabarmati river basin is shared with West Banas and Luni rivers, eastern boundary is formed by Banas and Mahi rivers. The southwestern boundary is an administrative border with Gujarat State. The basin extends over parts of Dungarpur, Pali, Sirohi and Udaipur Districts. The total catchment area of the Basin is 4,095.6 km².

Sabarmati River originates in the Aravali hills in Mount Abu area, and flows in a southwesterly direction. The main tributaries of the Sabarmati river are Wakal river and the Sel Nadi, which also originate in the same Aravali hill range west of Udaipur city and flow south-westwards in courses generally parallel to the Sabarmati river, up to their confluence with the river (in Gujarat).

Administrative Set-up:

Administratively, Sabarmati River Basin extends over parts of Dungarpur, Pali, Sirohi and Udaipur Districts encompassing 12 Blocks and 757 towns and villages.

S. No.	District Name	Area (sq km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Dungarpur	606.0	14.8	3	137
2	Pali	1.4	-	1	-
3	Sirohi	23.0	0.6	2	1
4	Udaipur	3,465.2	84.6	6	619
Total		4,095.6	100.0	12	757

Climate:

The Rajasthan part of Sabarmati River basin primarily falls within hilly areas of Aravalli Range and has a humid to sub-humid. It is extremely cold from October to February while turning hot from March to September. Overall, it receives good rainfall during monsoon period that extends during the four Monsoon months (June-September). The annual mean rainfall over the Sabarmati Basin was computed as 575 mm, of which about 96% falls during these four months as non-monsoon rainfall is negligible.

TOPOGRAPHY

SABARMATI RIVER BASIN

The major part of the Basin is marked by hilly terrain belonging to the Aravali chain marked by steep slopes and narrow valleys. The hills trend in northeast-southwest direction. The elevation variation within the part of basin is very high (to the extent of about 1000m). The higher elevations (about 1168m) have been observed in the northeastern portion and lower elevations (about 175m) are observed in the southern part of the basin indicating general sloping downwards towards south to southwestern direction. The river and its tributaries either flow parallel to the ridges in NW-SE direction or cut across them in a SE-NW or E-W directions.

Table: District wise minimum and maximum elevation

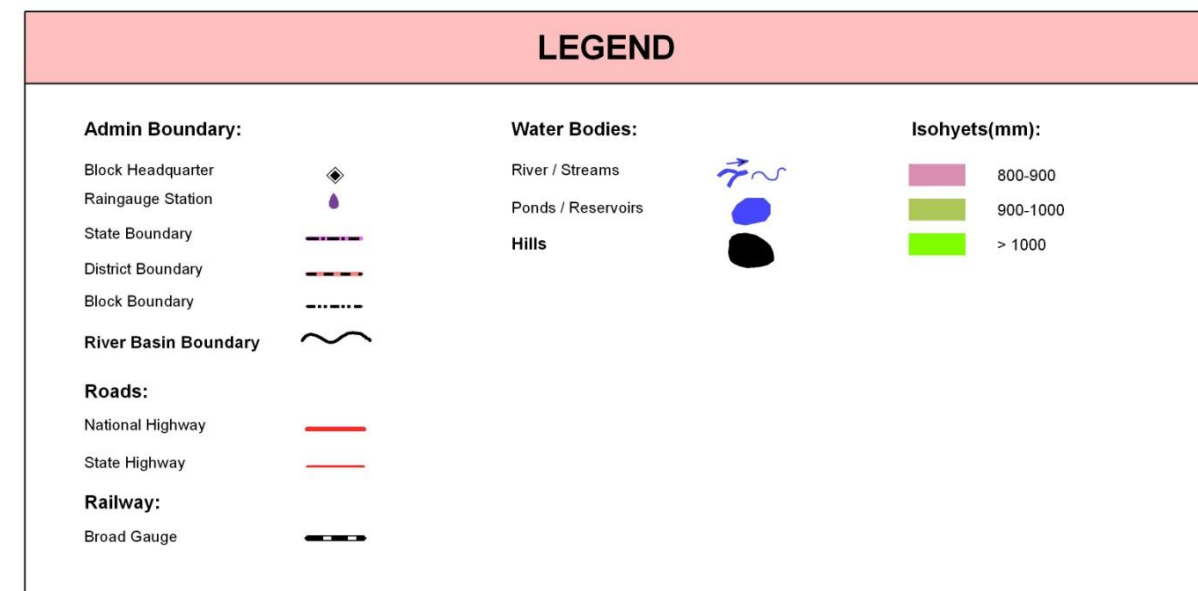
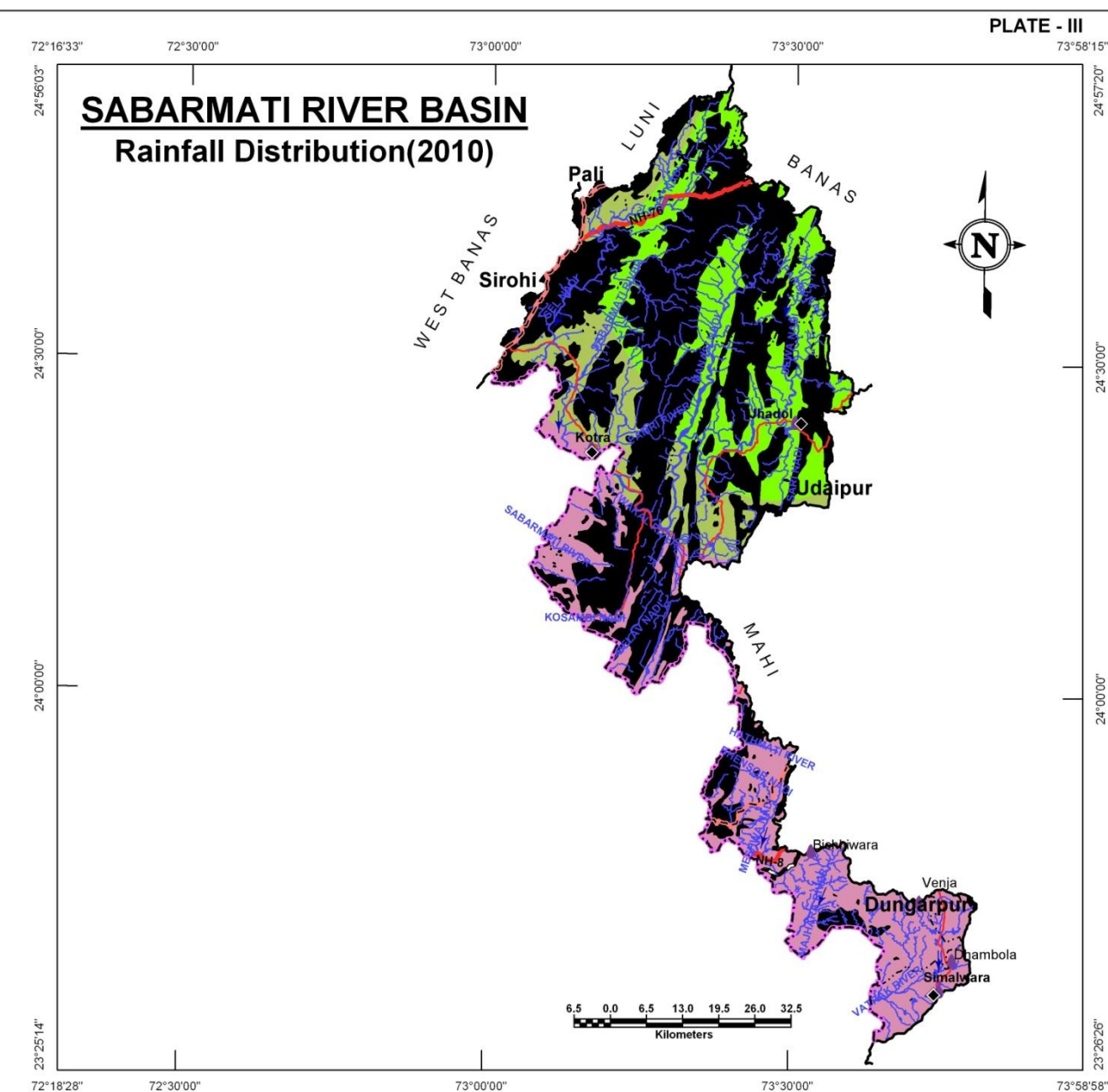
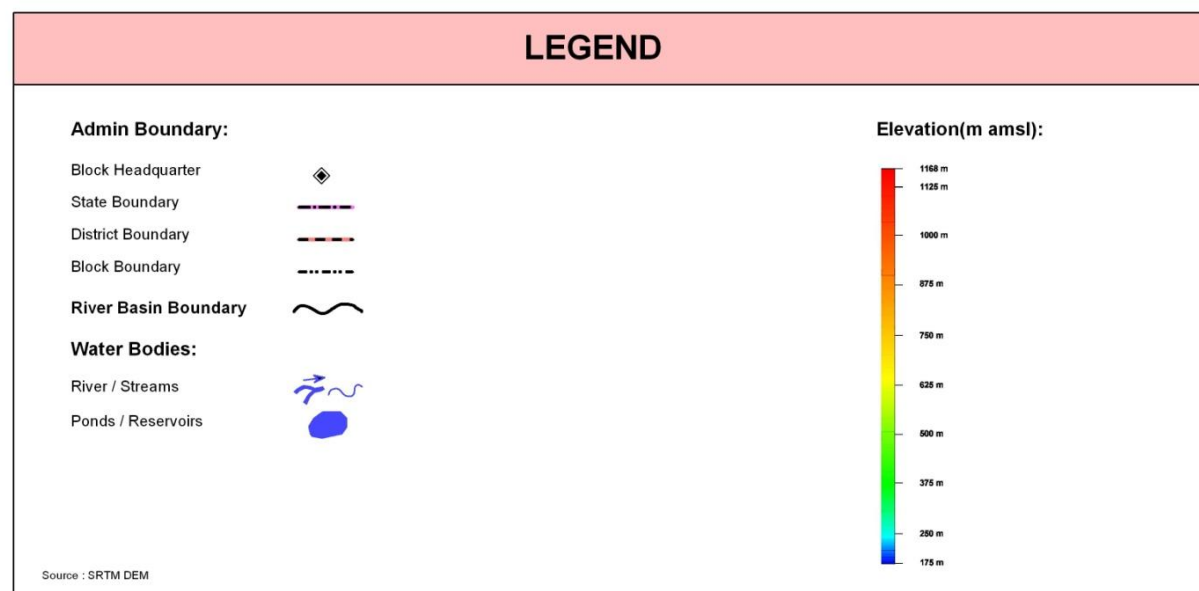
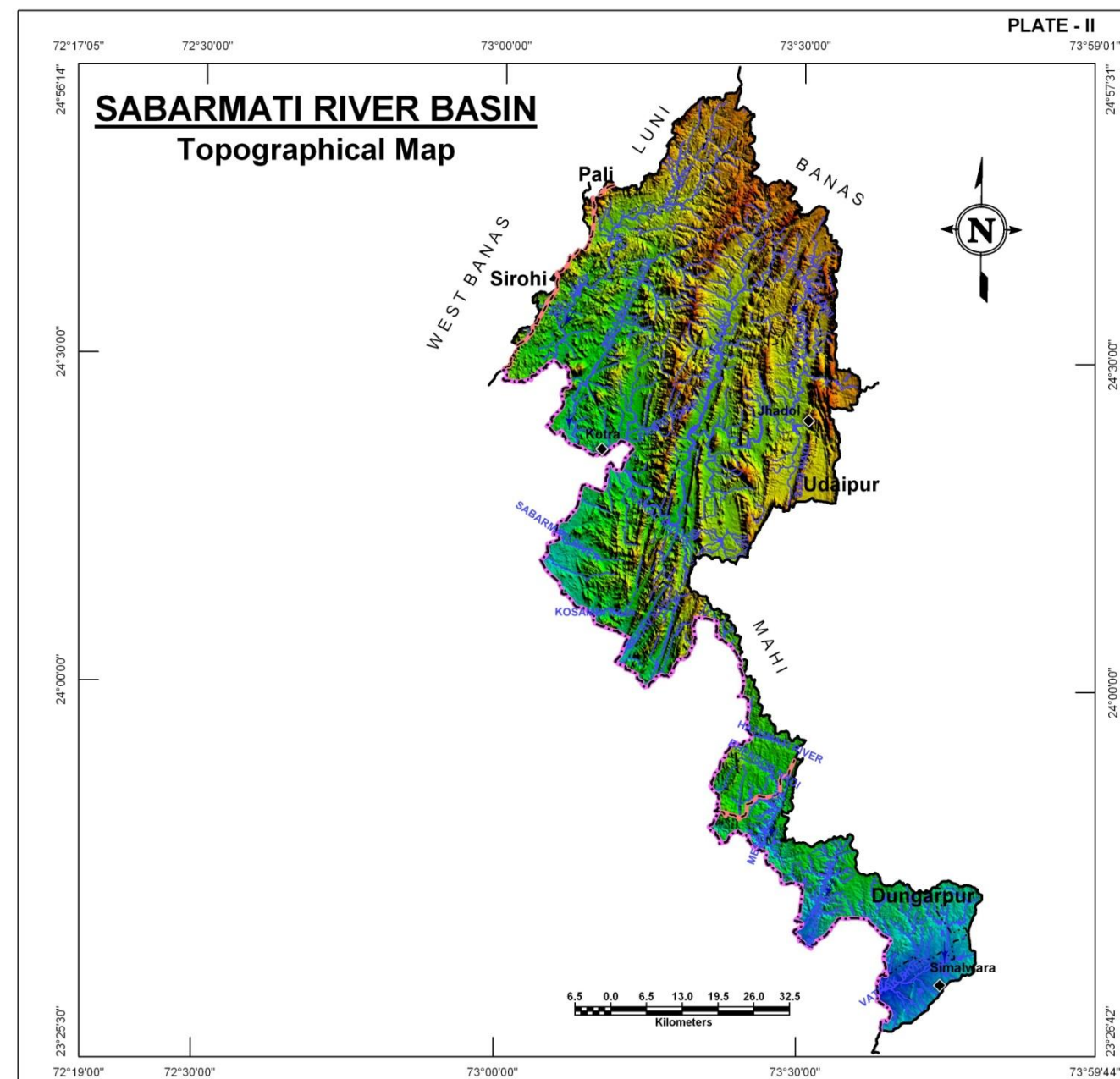
S. No.	District Name	Min Elevation (m amsl)	Max. Elevation (m amsl)
1	Dungarpur	174.6	496.9
2	Pali	683.8	952.2
3	Sirohi	434.3	901.7
4	Udaipur	242.1	1,167.0

RAINFALL

The area of the basin being small, hill and less inhabited, has very few rain gauge stations. The rainfall distribution map (Plate – III) reveals high rainfall areas being located in the hilly areas in Udaipur district and reduces further southeastwards reaching minimum in Dungarpur area. The total average rainfall in the year 2010 had been around 1000mm.

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajasthan.gov.in>))

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
1	Jhadol	1,081.0	84.0	1,165.0
2	Kotra	782.0	114.0	896.0



GEOLOGY

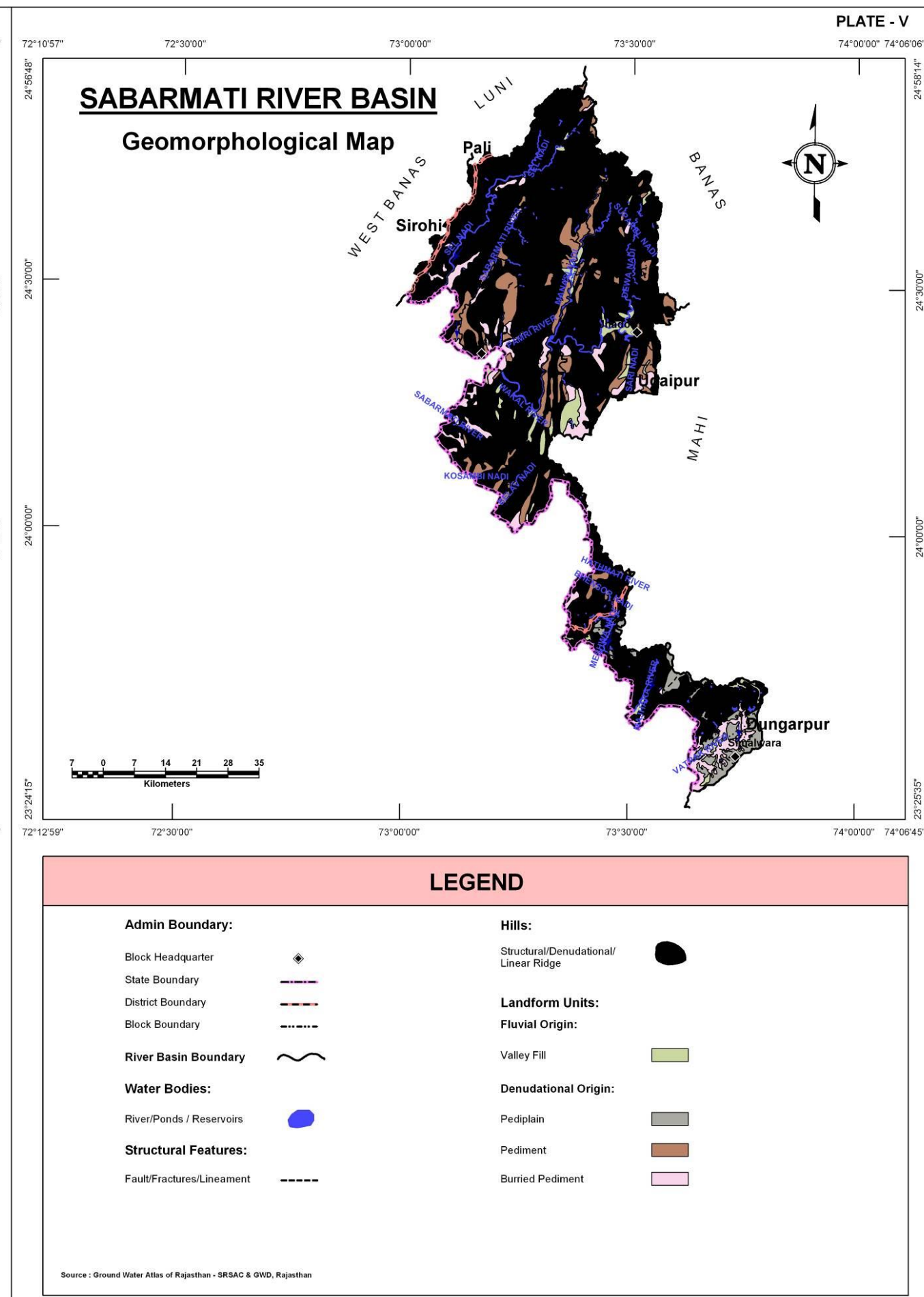
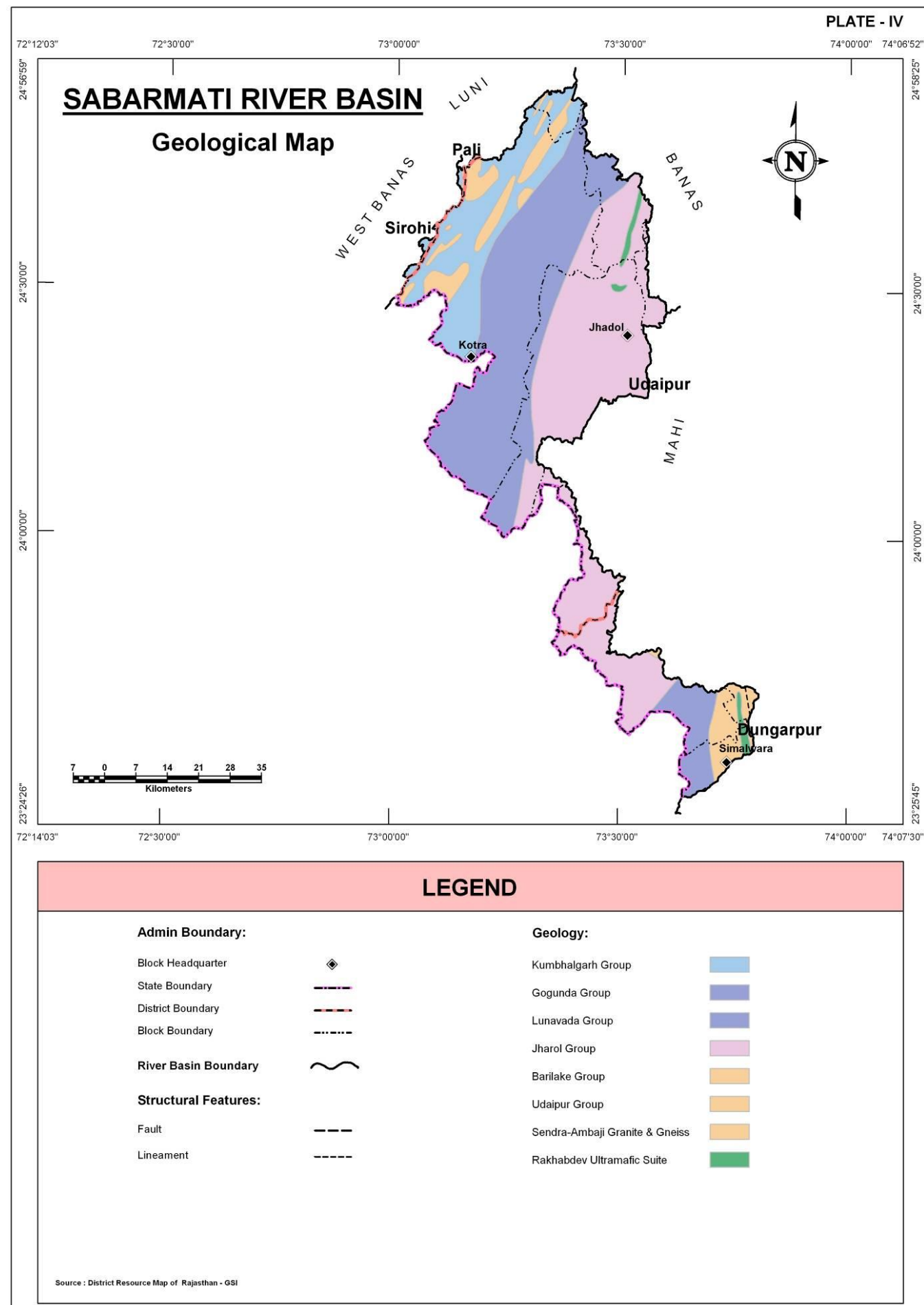
SABARMATI RIVER BASIN

The Sabarmati River Basin is covered mainly by rocks belonging to the Pre-Cambrian rocks (Delhi Super-Group and Aravalli Super - Group). Very limited area, i.e., valleys intervening hills has deposits of eolian and fluvial origin of Recent to Sub-Recent age.

Age	Super-Group	Group/Formation	Rock Types
Sub-Recent To Recent	Alluvium (and Colluvium)	--	Pebbles, gravel. Sand, silt, clay
-----X-----X----- Unconformity -----X-----X-----			
Lower Precambrian To Upper Precambrian	Delhi	Kumbhalgarh (with Sandra Ambaji Intrusives)	Calc-schists and gneisses, marble, mica-schists and Quartzites with Synorogenic Granites, gneisses, migmatites
		Gogunda	Quartzites, biotite- Schists, calc-schists, Hornblende schists, Calc-silicates
Lower Precambrian (Algonkian)	Aravalli	Champaner	Phyllites, meta proto Quartzites, meta Greywackes, Metaconglomerates, Dolomites, Mica-schists, etc.
		Lunavada	Phyllites, meta Siltstone, meta Quartzites, meta-sub Greywackes, Garnetiferous Mica-schists, meta Conglomerates, Dolomites
		Jharol	Chlorite phyllites, Quartzites, mica Schists, biolite Schists, etc.

GEOMORPHOLOGY

Origin	Landform Unit	Description
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
	Pediplain	Coalescence and extensive occurrence of pediment.
Fluvial	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



AQUIFERS

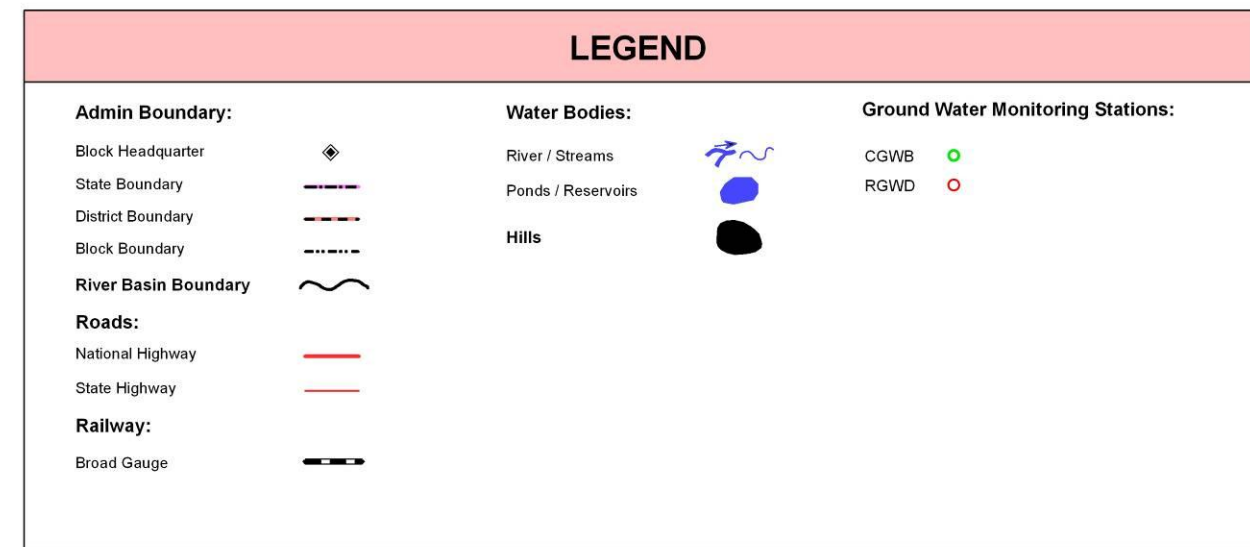
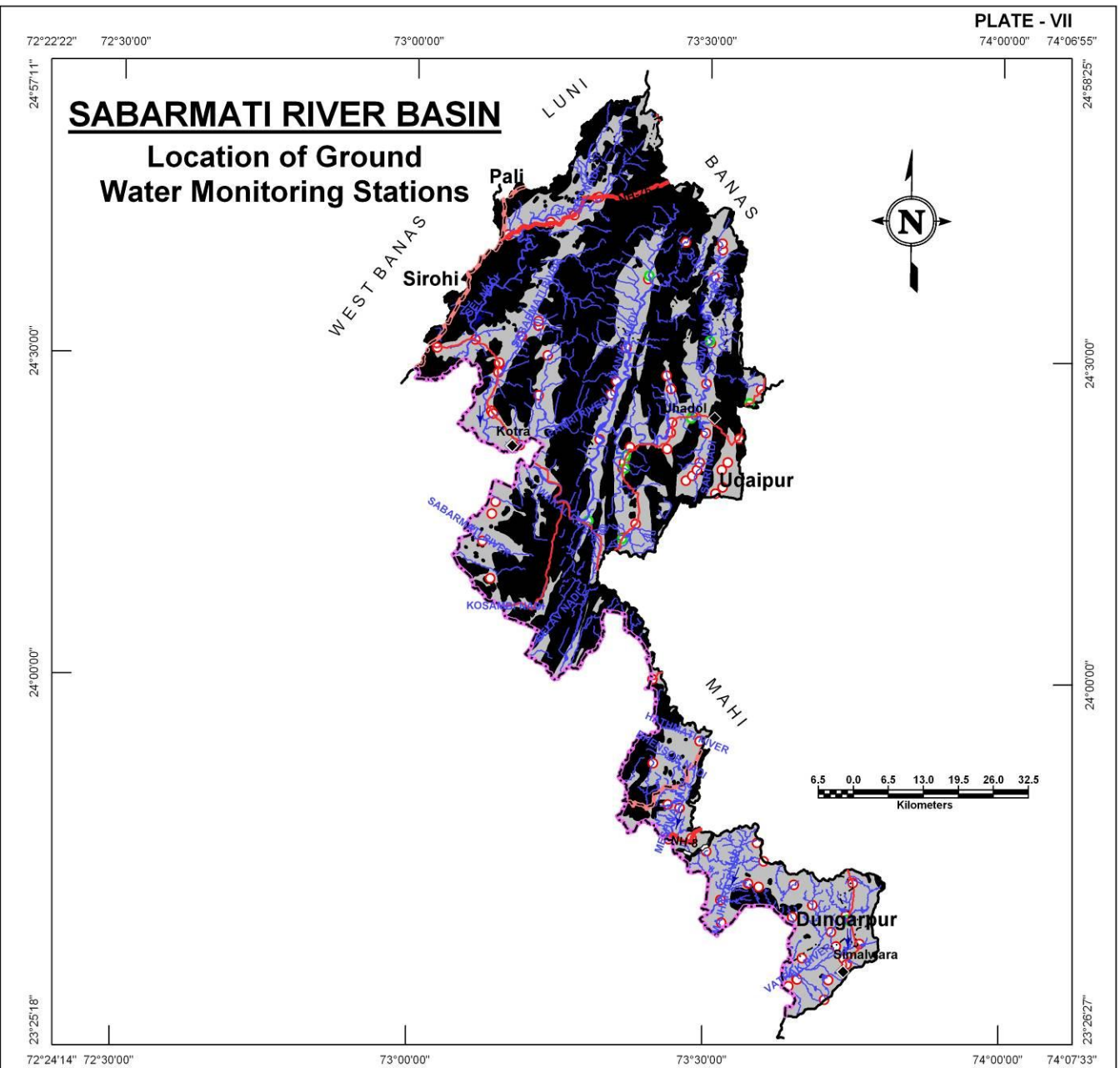
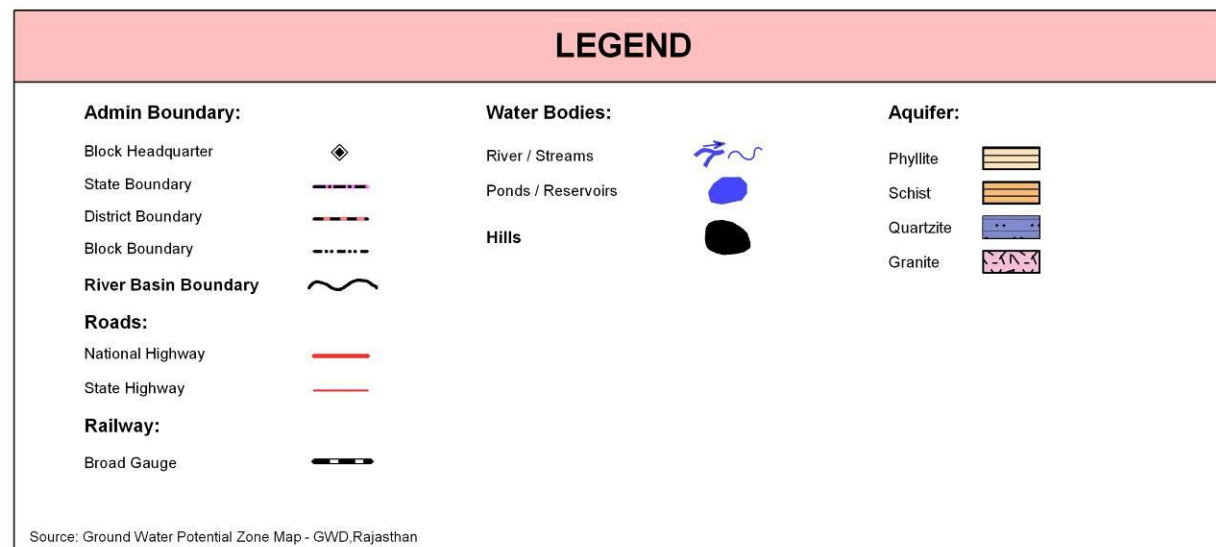
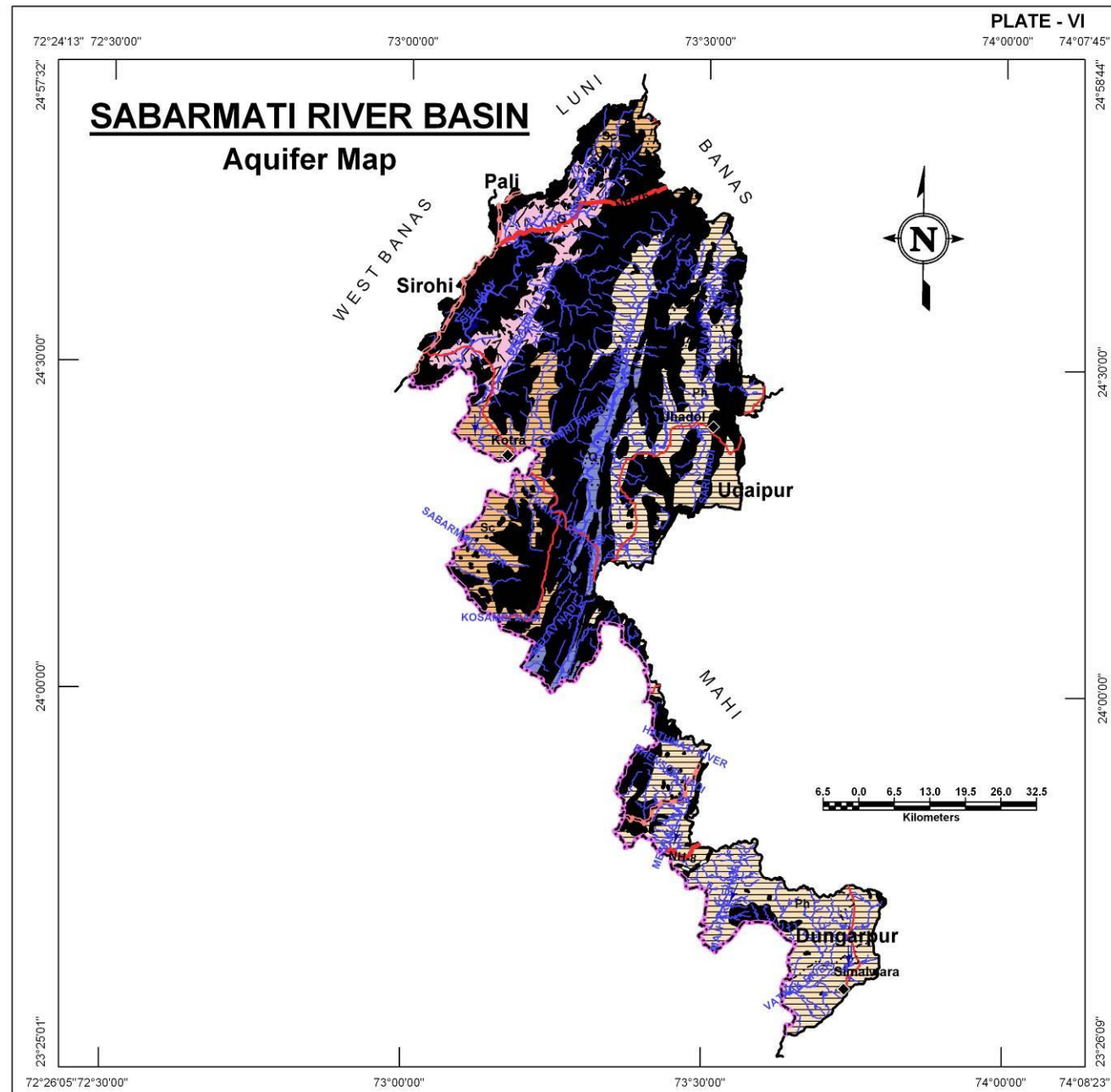
Phyllites constitute the predominant aquifer type as can be seen in Plate – VI, which occupy close to 30% of the basin area. This is followed by Schist aquifers (about 9% of area) and then by Granites and Quartzites. Weathering and fracturing of rocks within each lithologic type results into formation of aquifers within the massive hard rocks. It can be seen both from map and statistical table below which indicates the predominance of hilly area within the basin that occupy more than half of the basin area.

Aquifer in Potential Zone	Area (sq km)	% of Basin Area	Description of the unit/Occurrence
Phyllite	1,225.3	29.9	These include meta sediments and represented by carbonaceous phyllite.
Schist	354.4	8.7	Medium to fine grained compact rock. The ltho units are soft, friable and have closely spaced cleavage.
Quartzite	140.3	3.4	Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.
Granite	297.9	7.3	Light grey to pink colour, medium to coarse grained, and characteristically have porphyritic texture.
Non Potential Zone	2,077.7	50.7	Hills
Total	4,095.6	100.0	

LOCATION OF GROUND WATER MONITORING WELLS

The basin has a fairly well distributed network of ground water monitoring stations (103) in the basin owned by RGWD (93) and CGWB (10); and an additional one well has been recommended to be added to the water quality monitoring network to add more effectiveness to it while the existing wells are sufficient to monitor water level in the basin.

District Name	CGWB	RGWD	Total	Recommended additional wells for optimization of monitoring network	
				Water Level	Water Quality
Dungarpur	2	33	35	-	1
Pali	-	-	-	-	-
Sirohi	-	-	-	-	-
Udaipur	8	60	68	-	1
Total	10	93	103	0	1



LOCATION OF EXPLORATORY WELLS

In all, there are 36 exploratory wells present in the basin drilled in the past by RGWD (26) and CGWB (10) that form basis for delineation of sub-surface aquifers. Large part of the area being hilly terrain and by and large excluded from detailed exploration, the density of exploratory wells in rest of the area from ground water exploration perspective, is good.

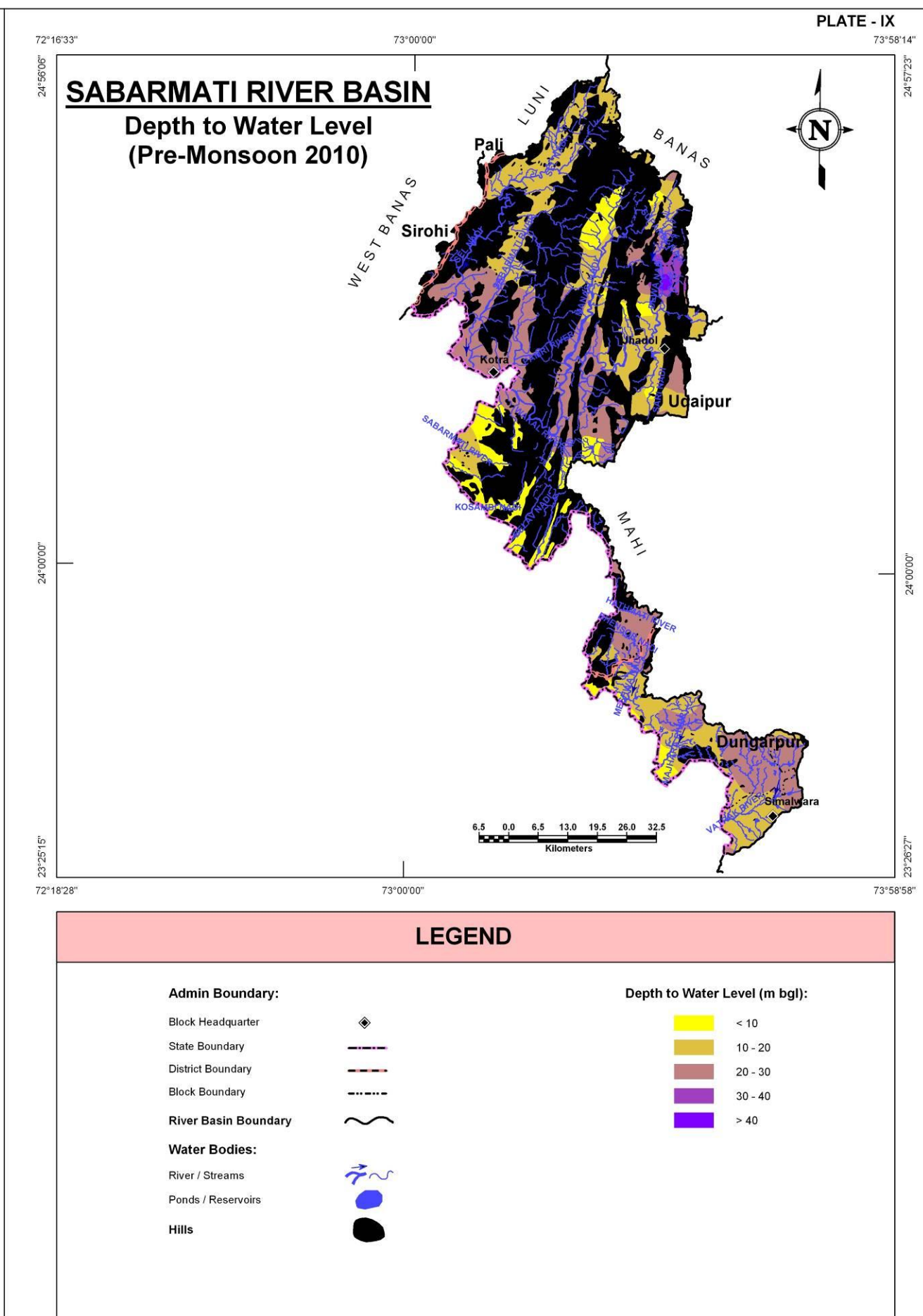
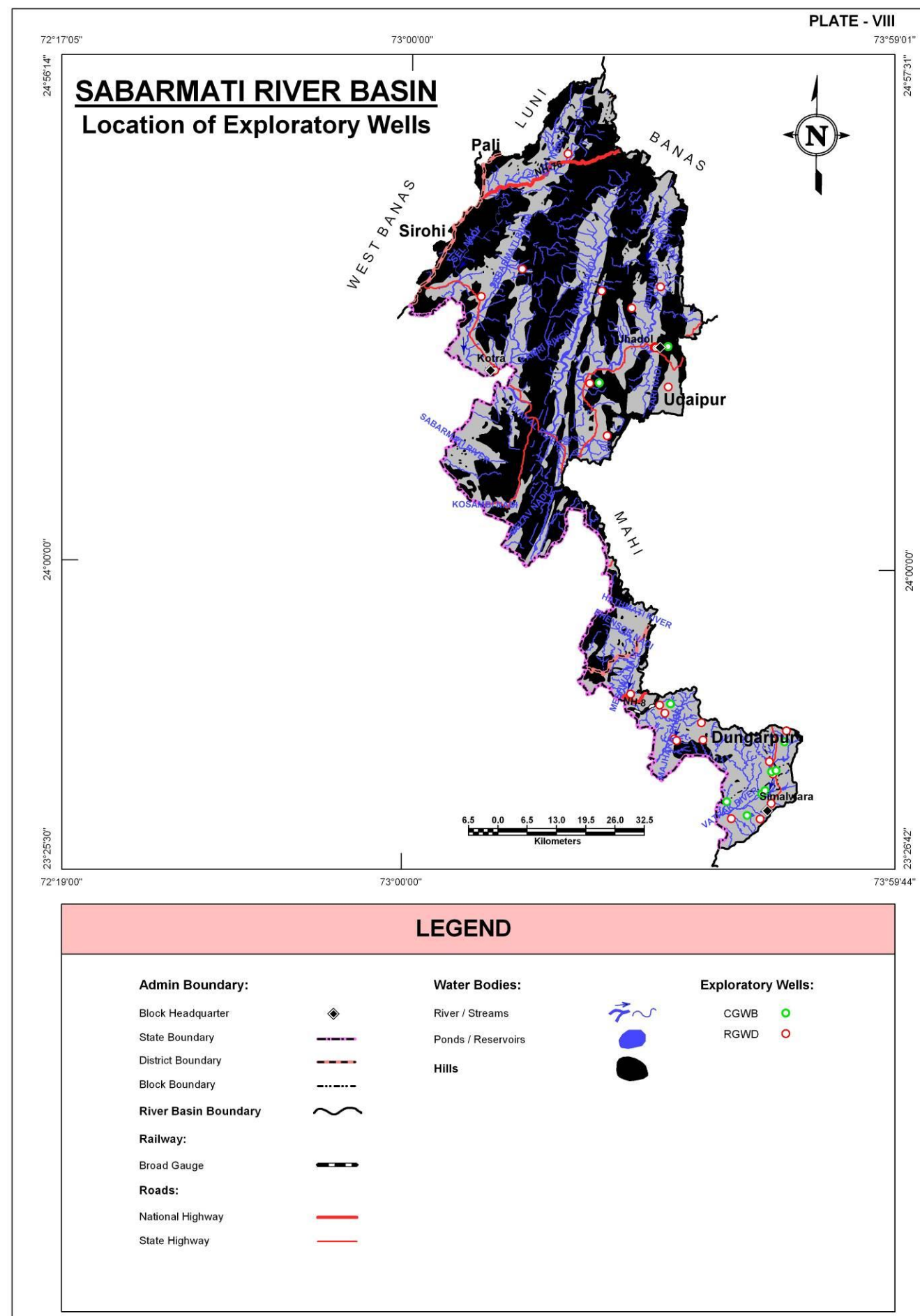
District Name	CGWB	RGWD	Total
Dungarpur	8	13	21
Pali	-	-	-
Sirohi	-	-	-
Udaipur	2	13	15
Total	10	26	36

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The general depth to water level in the basin ranges from 10 to 30 meters below ground level, as seen on northeastern, central, western and southwestern parts of the basin. These two interval ranges (10-20m and 20-30m bgl) together occupy more than 84% of the non-hilly basin area. Ground water at very shallow depths (<10m bgl) also occurs in large parts in Udaipur and Dungarpur districts covering an area of about 284 sq kms.

Depth to water level (m bgl) Pre Monsoon - 2010	District wise area coverage (sq km)*				Total Area (sq km)
	Dungarpur	Pali	Sirohi	Udaipur	
< 10	43.2	-	-	240.8	284.0
10 - 20	269.1	-	-	615.6	884.7
20 - 30	240.5	-	-	572.4	812.9
30 - 40	-	-	-	29.7	29.7
> 40	-	-	-	6.6	6.6
Total	552.8	-	-	1,465.1	2,017.9

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON 2010)

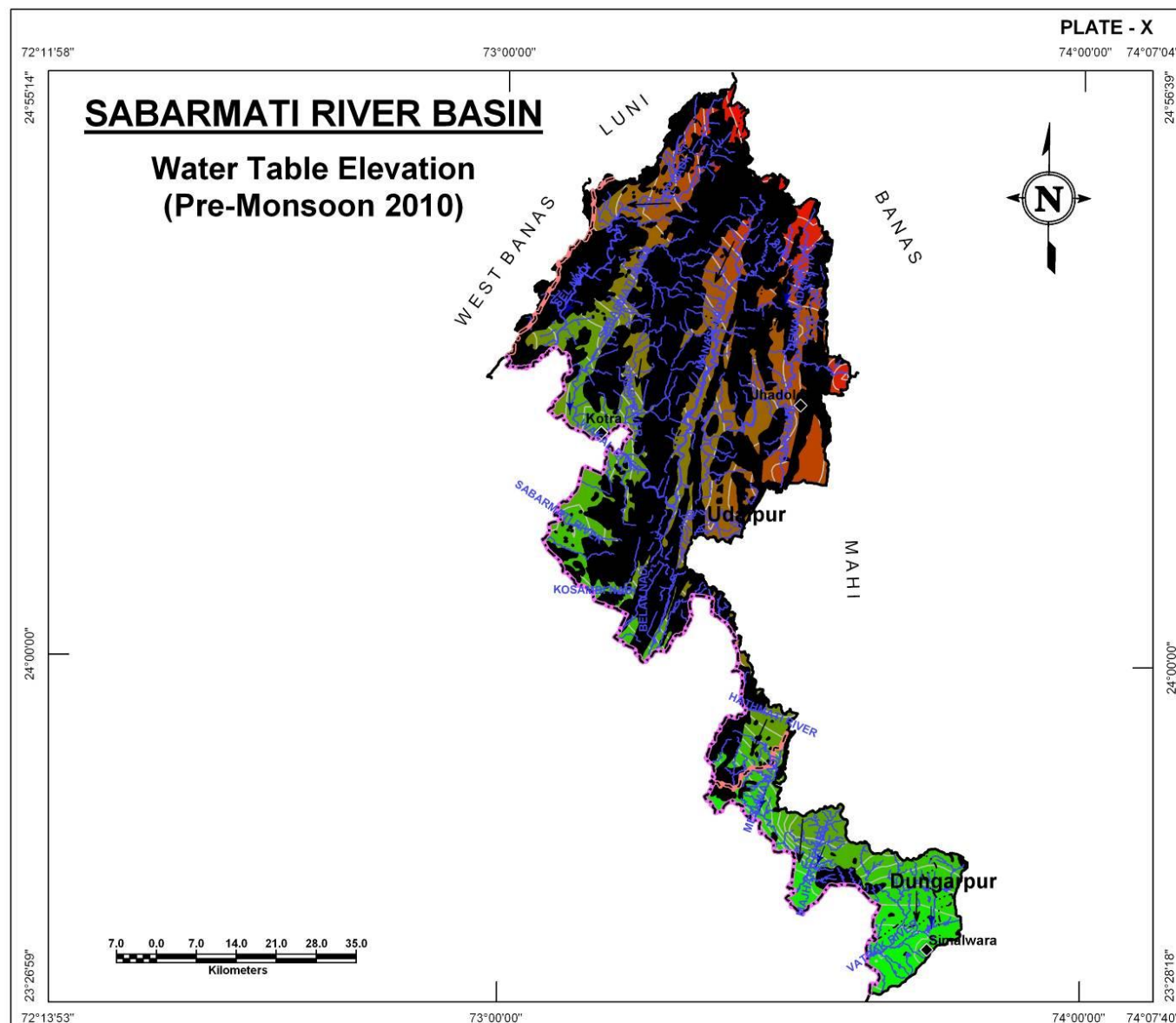
Water table elevation shows large variation since the topography in the basin reaches a high of about >1200m amsl and low of about 175m amsl. A perusal of Plate X reveals steep ground water flow gradients in the hilly areas of Udaipur district where the elevation ranges from about 840m amsl to about 400m amsl at the marginal areas of hills in the western part. The initial flow direction is from northeast to southwest till it reaches the plain areas west of Aravallis. The nearly isolated basin part of Dungarpur district shows a relatively lower water table elevation ranges below 400m amsl to the lowest of the basin of about 180m amsl coupled with lower flow gradients.

Water Table Elevation (m amsl) Pre Monsoon - 2010	District wise area coverage (sq km)				Total Area (sq km)
	Dungarpur	Pali	Sirohi	Udaipur	
< 180	0.6	-	-	-	0.6
180 - 200	80.9	-	-	-	80.9
200 - 220	114.7	-	-	-	114.7
220 - 240	79.5	-	-	2.0	81.5
240 - 260	65.9	-	-	4.9	70.8
260 - 280	66.5	-	-	21.5	88.0
280 - 300	39.6	-	-	74.3	113.9
300 - 320	51.6	-	-	100.1	151.7
320 - 340	52.9	-	-	93.9	146.8
340 - 360	0.6	-	-	106.6	107.2
360 - 380	-	-	-	58.4	58.4
380 - 400	-	-	-	42.8	42.8
400 - 440	-	-	-	108.5	108.5
440 - 480	-	-	-	116.2	116.2
480 - 520	-	-	-	191.5	191.5
520 - 560	-	-	-	179.5	179.5
560 - 600	-	-	-	136.3	136.3
600 - 640	-	-	-	94.0	94.0
640 - 680	-	-	-	27.2	27.2
680 - 720	-	-	-	33.5	33.5
720 - 760	-	-	-	34.0	34.0
760 - 800	-	-	-	20.2	20.2
800 - 840	-	-	-	18.0	18.0
> 840	-	-	-	1.7	1.7
Total	552.8	-	-	1,465.1	2,017.9

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

The Basin has predominantly hardrock aquifers constituting most of the non-hilly parts. Therefore, wide fluctuations in water levels are seen to rise by very high numbers to the extent of 14m in post-monsoon period with respect to pre-monsoon water levels. Apart from very high fluctuation areas that are limited in spatial distribution, the general variation is observed in the range of 2m to 10m.

District Name	District wise area coverage (sq km) within fluctuation range								Total Area (sq km)
	< 2	2 to 4	4 to 6	6 to 8	8 to 10	10 to 12	12 to 14	> 14	
Dungarpur	17.6	167.3	165.7	85.8	84.5	21.5	7.1	3.3	552.8
Pali	-	-	-	-	-	-	-	-	-
Sirohi	-	-	-	-	-	-	-	-	-
Udaipur	15.6	411.4	435.2	397.1	179.1	24.0	2.7	-	1,465.1
Total	33.2	578.7	600.9	482.9	263.6	45.5	9.8	3.3	2,017.9



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

District Boundary

Block Boundary

RiverBasin Boundary

Water Bodies:

River / Streams

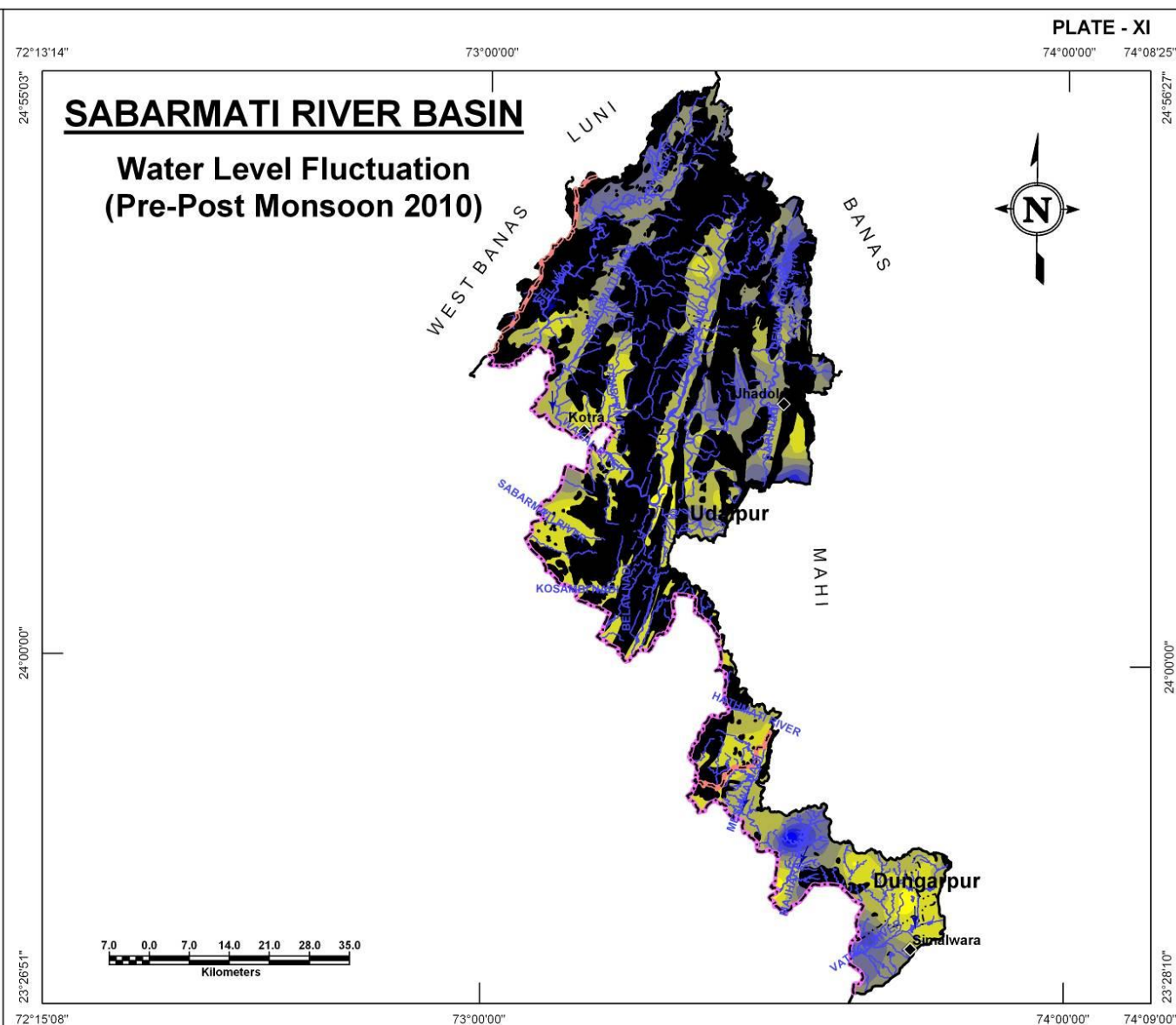
Ponds / Reservoirs

Hills

Water Table Elevation

Water Table Elevation(m amsl):

<180	400 - 440
180 - 200	440 - 480
200 - 220	480 - 520
220 - 240	520 - 560
240 - 260	560 - 600
260 - 280	600 - 640
280 - 300	640 - 680
300 - 320	680 - 720
320 - 340	720 - 760
340 - 360	760 - 800
360 - 380	800 - 840
380 - 400	>840



LEGEND

Admin Boundary:

Block Headquarter

State Boundary

District Boundary

Block Boundary

River Basin Boundary

Water Bodies:

River / Streams

Ponds / Reservoirs

Hills

Water Level Fluctuation(m):

< 2
2 to 4
4 to 6
6 to 8
8 to 10
10 to 12
12 to 14
>14

ELECTRICAL CONDUCTIVITY DISTRIBUTION

SABARMATI RIVER BASIN

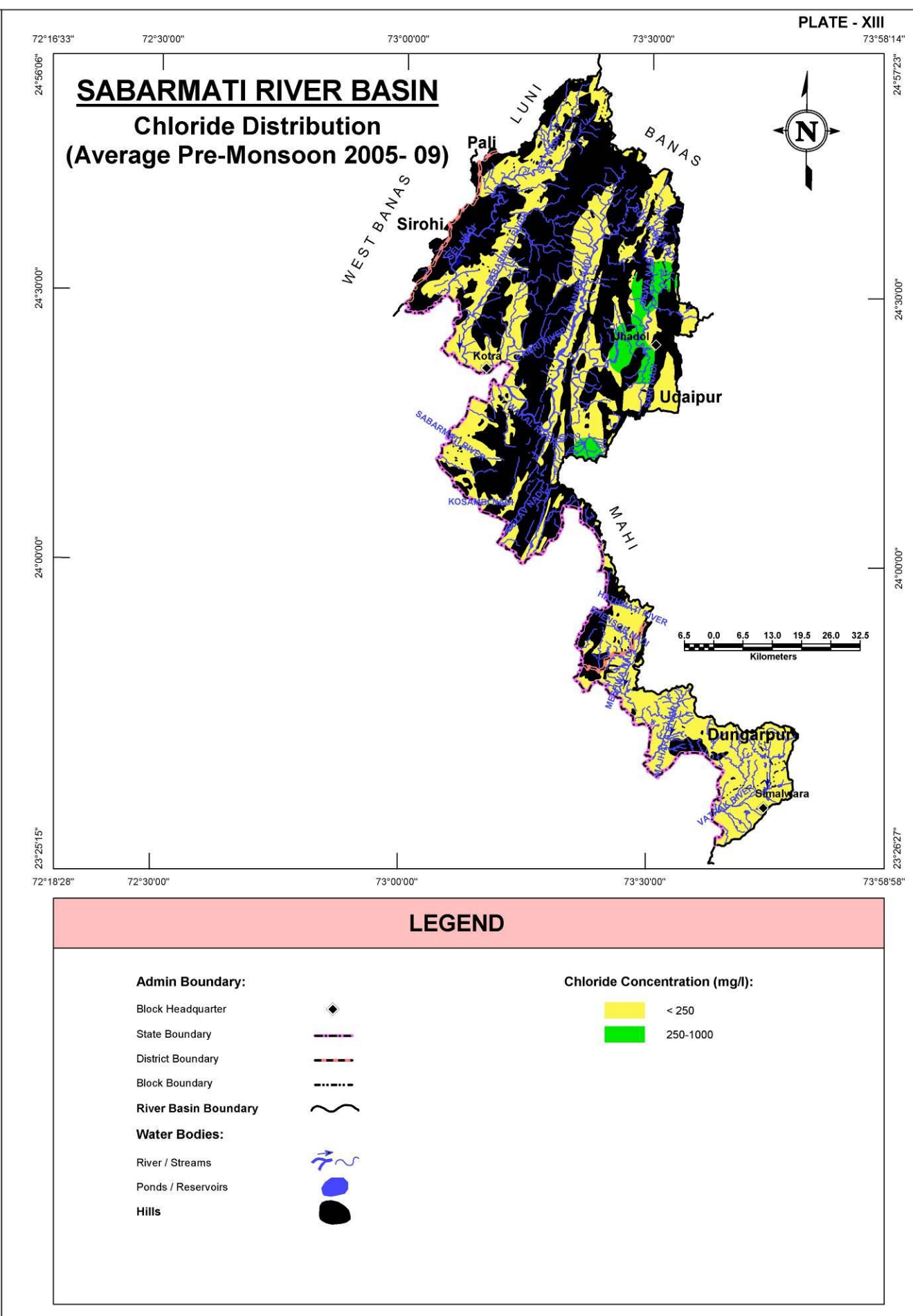
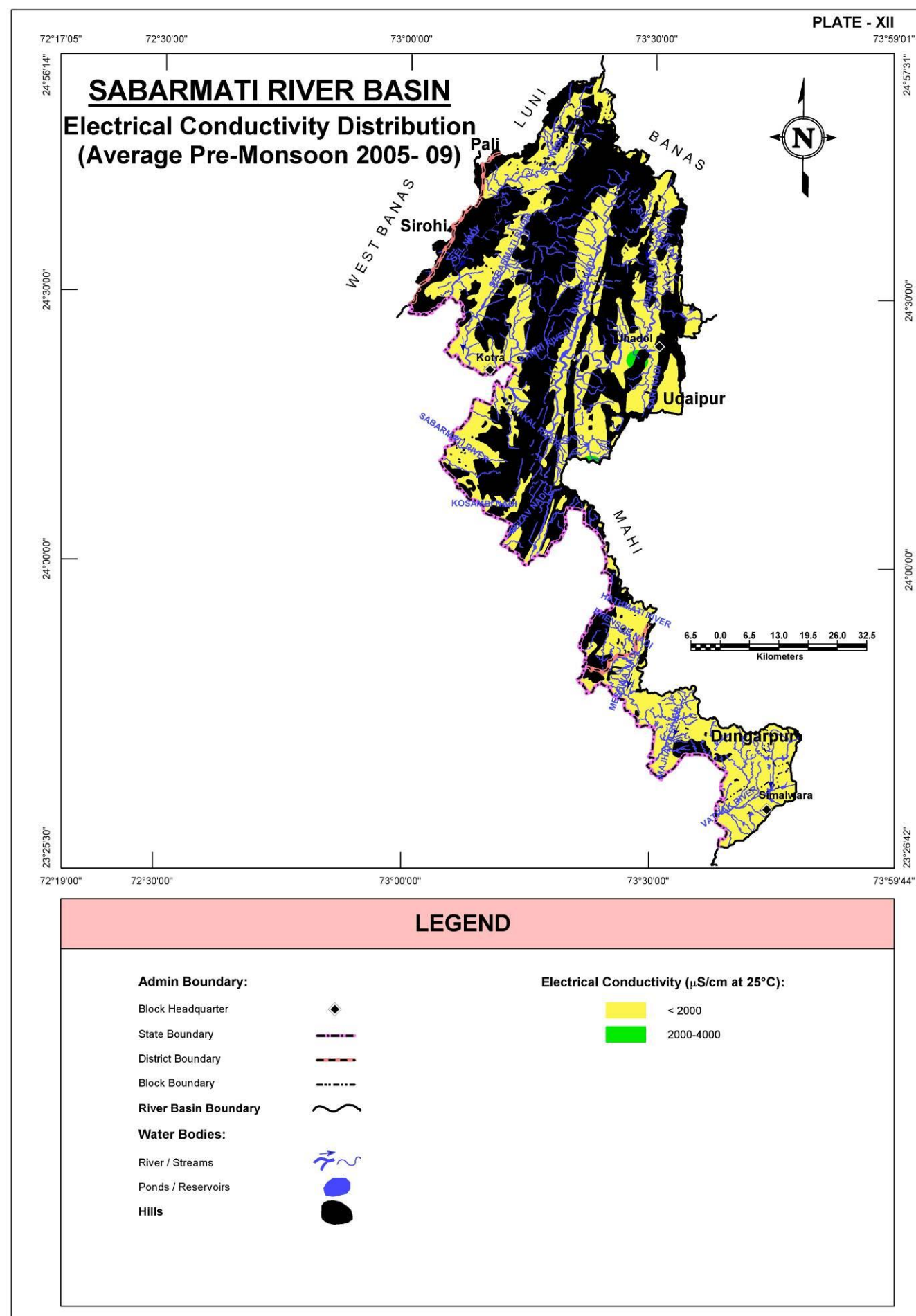
The Electrical Conductivity (at 25°C) distribution map is presented in Plate XII reveals that almost the entire basin has good quality of water suitable for domestic and other purposes. Approximately 99.5% of the non-hilly basin area has shown low EC values i.e., falling within the < 2000 $\mu\text{S}/\text{cm}$ range. Very small area (0.5% of the basin) around Jhadol in Udaipur district has shown EC values in the range of 2000-4000 $\mu\text{S}/\text{cm}$. There is no area that has high EC values of more than 4000 $\mu\text{S}/\text{cm}$.

Electrical Conductivity Ranges (µS/cm at 25°C) (Ave. of years 2005-09)	District wise area coverage (sq km)								Total Area (sq km)
	Dungarpur		Pali		Sirohi		Udaipur		
	Area	% age	Area	% age	Area	% age	Area	% age	
< 2000	552.8	100.0	-	-	-	-	1,454.8	99.3	2,007.6
2000-4000	-	-	-	-	-	-	10.3	0.7	10.3
Total	552.8	100.0	-	-	-	-	1,465.1	100.0	2,017.9

CHLORIDE DISTRIBUTION

Chloride concentration distribution map in Plate XIII indicates that most of the Sabarmati river basin falls within the < 250 mg/l category that renders the ground water suitable for domestic and other purposes. About 93% of the basin area is covered by Yellow coloured region that leaves only 7% area with Green colour where the ground water falls within the 250-1000 mg/l category which still is within tolerable limits for domestic use. No area has shown very high chloride concentration (i.e., >1000mg/l).

Chloride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)								Total Area (sq km)
	Dungarpur		Pali		Sirohi		Udaipur		
	Area	% age	Area	% age	Area	% age	Area	% age	
< 250	551.5	99.8	-	-	-	-	1,318.7	90.0	1,870.2
250 - 1000	1.3	0.2	-	-	-	-	146.4	10.0	147.7
Total	552.8	100.0	-	-	-	-	1,465.1	100.0	2,017.9



FLUORIDE DISTRIBUTION

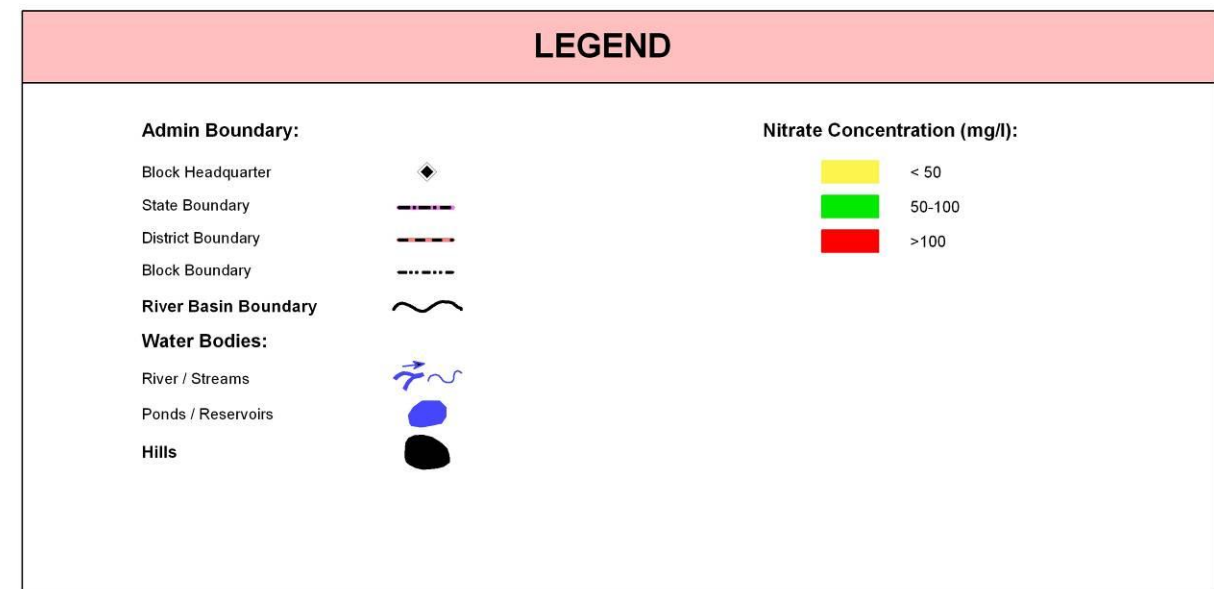
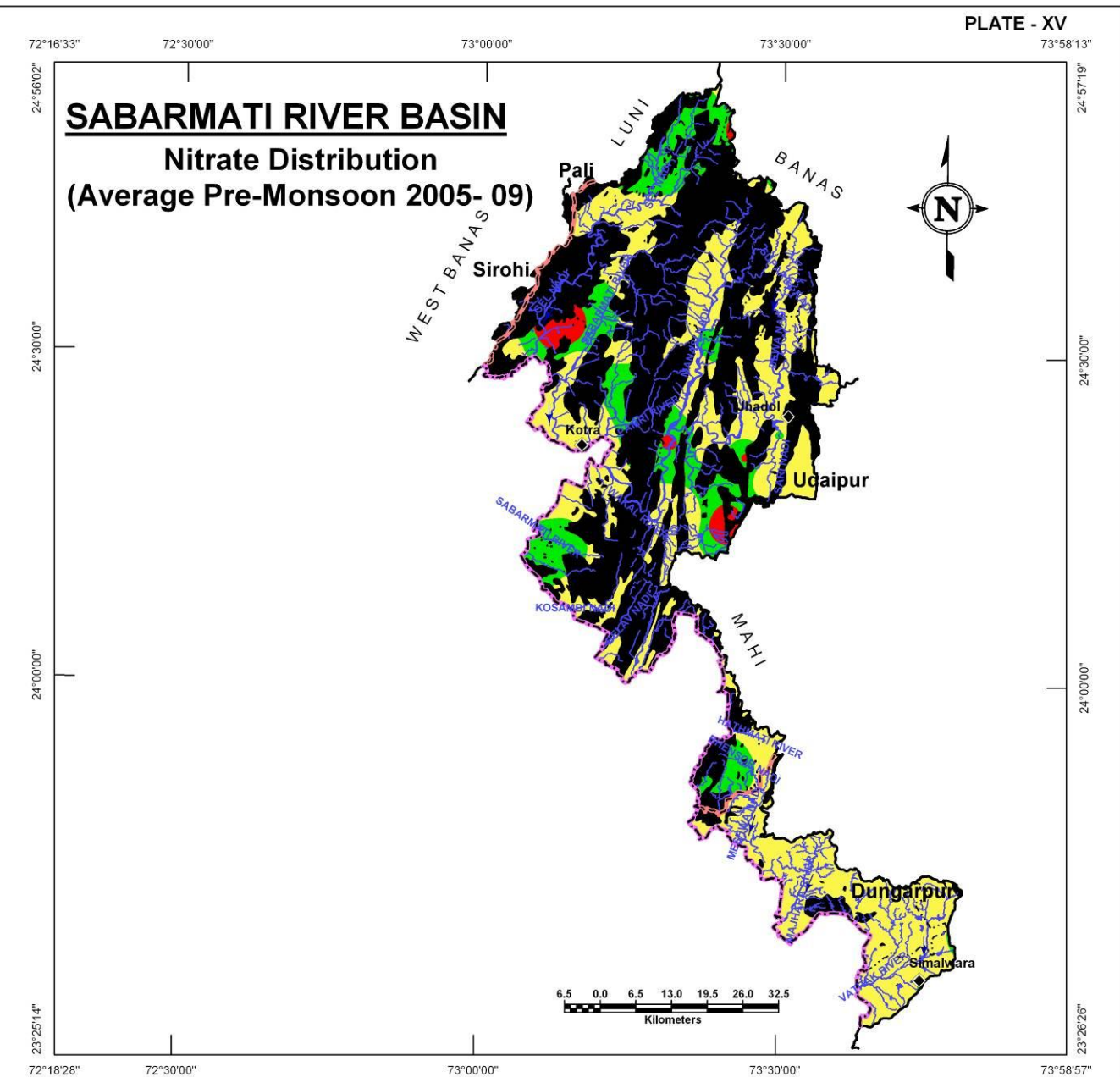
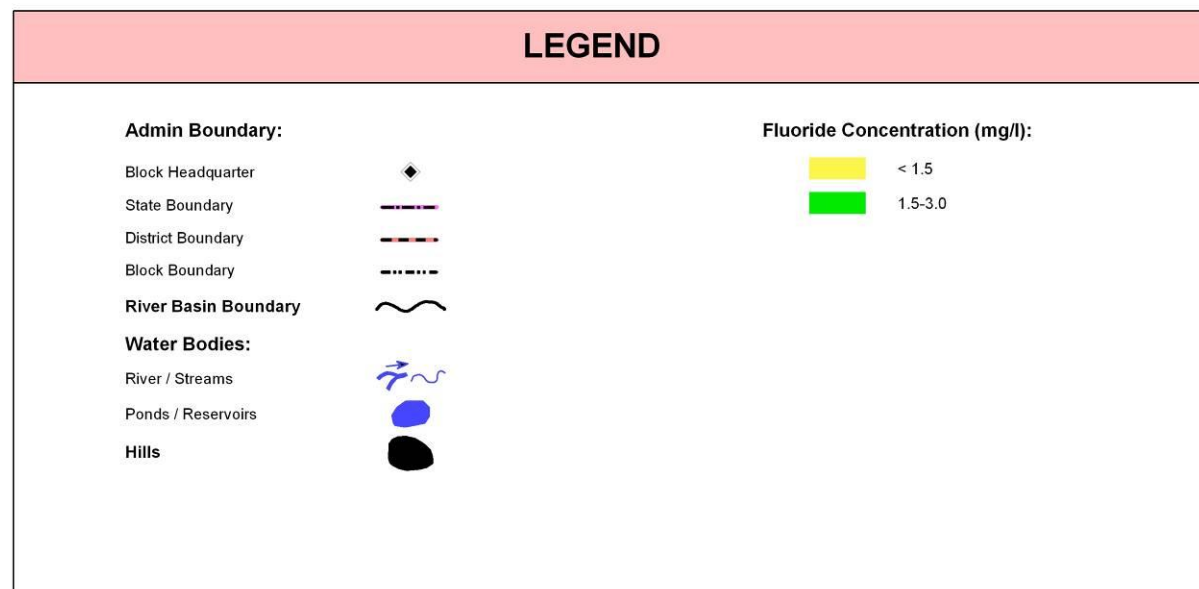
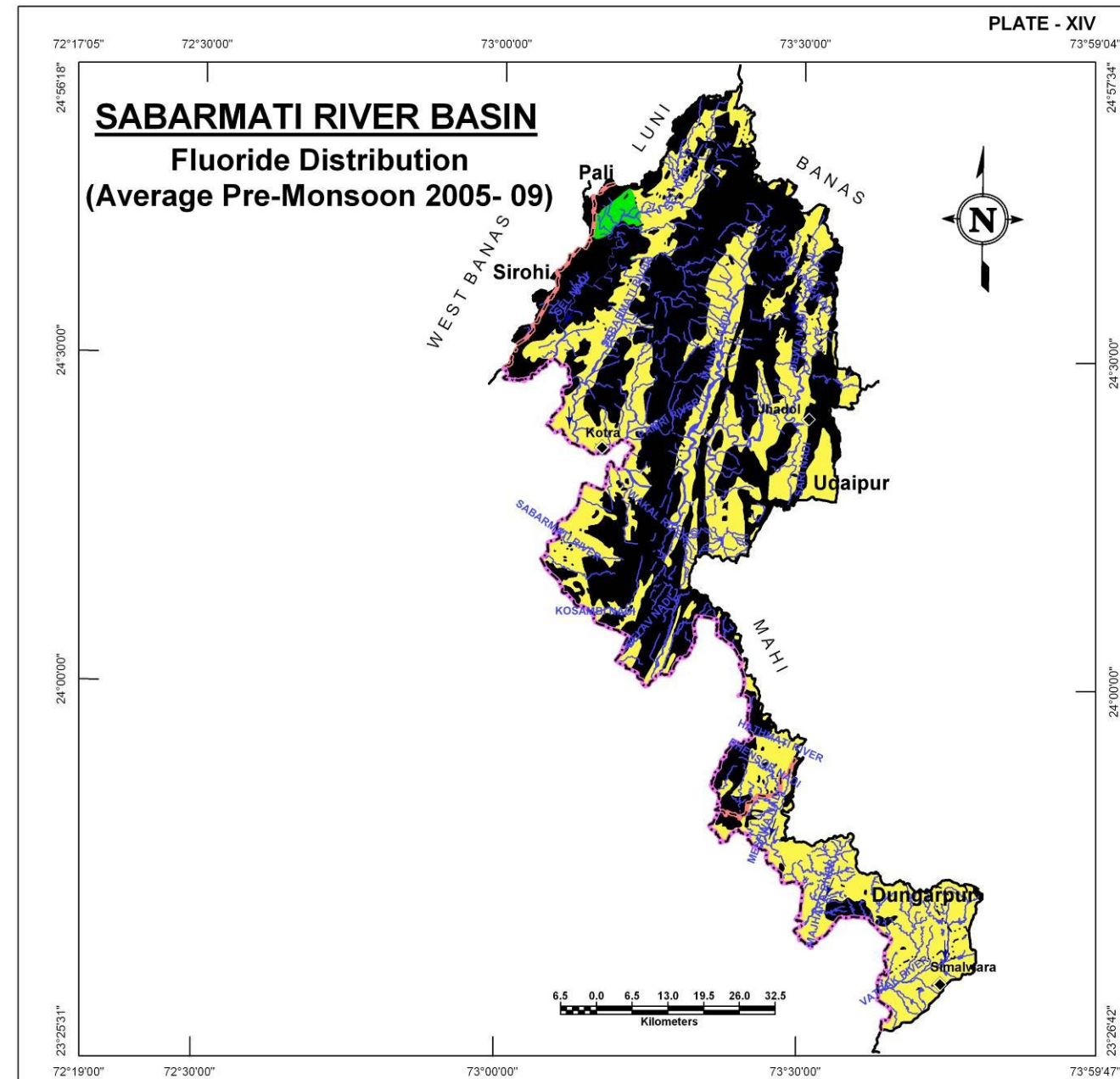
The Fluoride distribution map is presented in Plate XIV shows a very small patch (<2% of the basin area) in the northwestern part of the basin within Udaipur district near the Pali district border that has moderately high Fluoride concentration (1.5 – 3.0 mg/l). Remaining part of the basin which is approximately 98% of the basin area (excluding hilly area), falls within low fluoride concentration region which is suitable for domestic and other purposes.

Fluoride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)								Total Area (sq km)
	Dungarpur		Pali		Sirohi		Udaipur		
	Area	% age	Area	% age	Area	% age	Area	% age	
< 1.5	552.8	100.0	-	-	-	-	1,427.6	97.4	1,980.4
1.5-3.0	-	-	-	-	-	-	37.5	2.6	37.5
Total	552.8	100.0	-	-	-	-	1,465.1	100.0	2,017.9

NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate XV shows distribution of Nitrate in ground water. There are some isolated patches in the central and northern part of the basin that have shown high concentration of Nitrate in ground water (>100mg/l) and together these account for hardly 3% of the basin area, all within Udaipur district. Surrounding these high Nitrate concentration areas are moderate concentration areas (50-100 mg/l) covering about 21% of basin area. The rest of the basin area (excluding hilly areas) falls within low Nitrate concentration area where the ground water is suitable for agriculture.

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)								Total Area (sq km)
	Dungarpur		Pali		Sirohi		Udaipur		
	Area	% age	Area	% age	Area	% age	Area	% age	
< 50	545.9	98.7	-	-	-	-	978.4	66.8	1,524.3
50-100	6.9	1.3	-	-	-	-	426.1	29.1	433.0
> 100	-	-	-	-	-	-	60.6	4.1	60.6
Total	552.8	100.0	-	-	-	-	1,465.1	100.0	2,017.9



DEPTH TO BEDROCK

The entire area of the Sabarmati River basin is underlined by the hard rocks at different depths. The major rocks types occurring are Siltstone, shale, greywackes in sedimentary rock type, Schist, Gneiss, Quartzite, Slate, Phyllite in Metamorphic rock and migmatites in igneous rocks. These rocks are overlain by thin alluvial sands of rivers and streams and residual soils from the weathering of hard rocks. The depth to bed rock defines the sub surface topography of the occurrence of massive hard rock beneath alluvial deposits and weathered and fractured zone in hard rocks.

Perusal of the map (Plate – XVI) of depth to bed rock in meters above ground level reveals that the depth to bed rock is at shallow levels in northern part of the river basin which gradually deepens in south and southern part. The depth to bedrock in the river basin varies from 40-60 m bgl in the south, to less than 20m bgl in the northern parts. The general depth to bedrock has been in the 20-40m bgl range as about 62% of the basin area falls within this category.

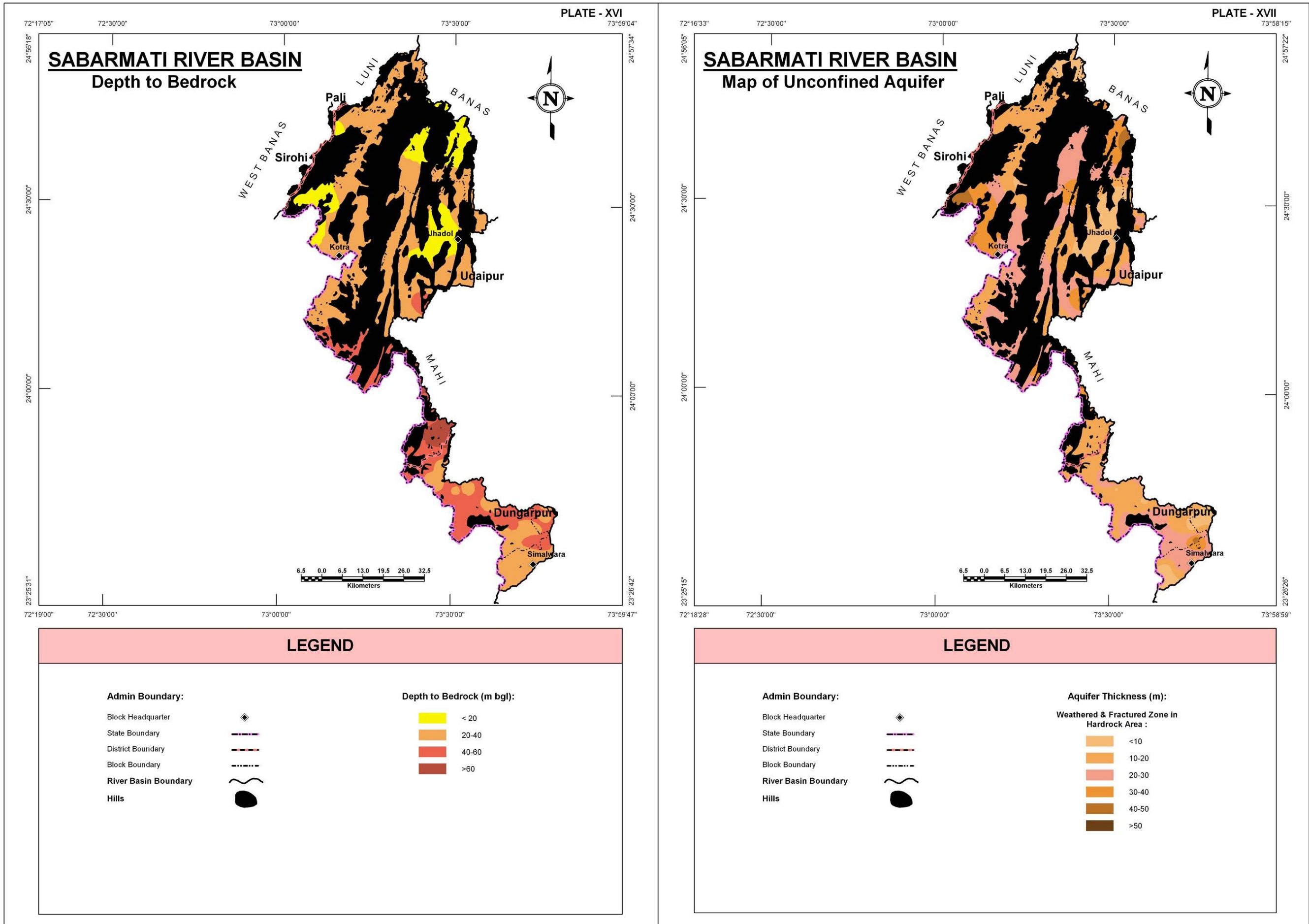
Depth to Bedrock (m bgl)	District wise area coverage (sq km)				Total Area (sq km)
	Dungarpur	Pali	Sirohi	Udaipur	
< 20	-	-	-	299.4	299.4
20-40	265.2	-	-	989.2	1,254.4
40-60	283.9	-	-	120.8	404.7
> 60	3.7	-	-	55.7	59.4
Total	552.8	-	-	1,465.1	2,017.9

UNCONFINED AQUIFER

Hardrock areas:

Weathered, fractured and jointed secondary openings within hardrock formations constitute aquifer within them in the basin. Perusal of Plate – XVII reveals that the 10m to 30m thickness zones occupies significant parts of the basin amounting to approximately 75% of the potential zone. The thicker aquifer (>30m) are mostly present in Udaipur district and at some places even reaching >50m but the latter however is very limited in extent. About 13% of the potential zones has very thin (<10m) horizon of aquifer.

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)				Total Area (sq km)
	Dungarpur	Pali	Sirohi	Udaipur	
<10	70.5	-	-	183.0	253.5
10-20	269.1	-	-	554.1	823.2
20-30	190.7	-	-	486.8	677.5
30-40	20.0	-	-	199.7	219.7
40-50	2.5	-	-	41.3	43.8
> 50	-	-	-	0.2	0.2
Total	552.8	-	-	1,465.1	2,017.9

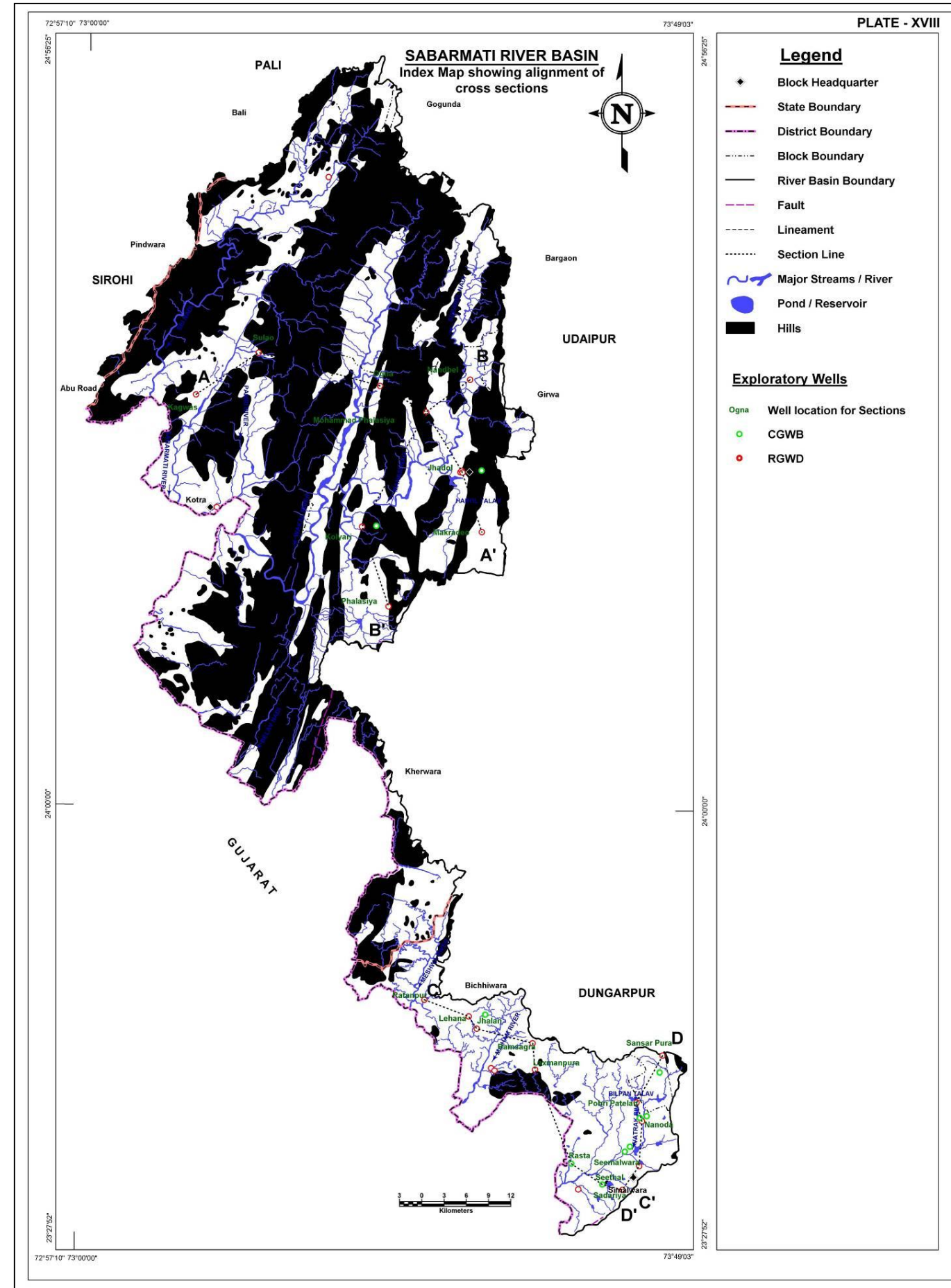


CROSS SECTIONS

SABARMATI RIVER BASIN

Several hydrogeological cross sections have been drawn to better depict the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their effect on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XXII. The broad alignment of the sections is as given below:

Name of Section Line	Orientation
Section AA'	W – E – SE
Section BB'	SW – NE
Section CC'	NW – SE
Section DD'	N – S



CROSS SECTIONS

SABARMATI RIVER BASIN

Section A-A':

The A-A' section (Plate – XIX) trends in W-E-SE direction extending to a distance of about 52.5km across the Sabarmati, Wakal and Manasi Rivers in the basin. The section depicts the disposition of different layers of gneiss, schist and phyllite overlain by top soil. The lithologs of 5 boreholes along with surrounding well information is taken while preparing the section. At a distance of about 20 km of 'A', there is a fault close to Manasi River where Gneisses (in the west) are juxtaposed against Schists (in the east)

The water level varies from 375 m amsl to 640 m amsl following the surface topography. The gradient is very steep in the eastern part which flattens out in the middle of the section and again becoming steep towards west.

Section B-B':

The section B-B' (Plate – XX) is trending SW – NE direction and covers an approximate length of 37.5 km, roughly parallel to Manasi River. The lithologs of 4 boreholes along with surrounding well information has been considered while preparing the section. The section reveals a predominantly schistose lithology throughout, except for a small patch of Phyllite in the southwestern part. Topography is high in the two ends of the section and low but wide in the centre.

Ground water level varies from 590 m amsl to 790 m amsl in this profile, parallel to surface topography.

PLATE - XIX

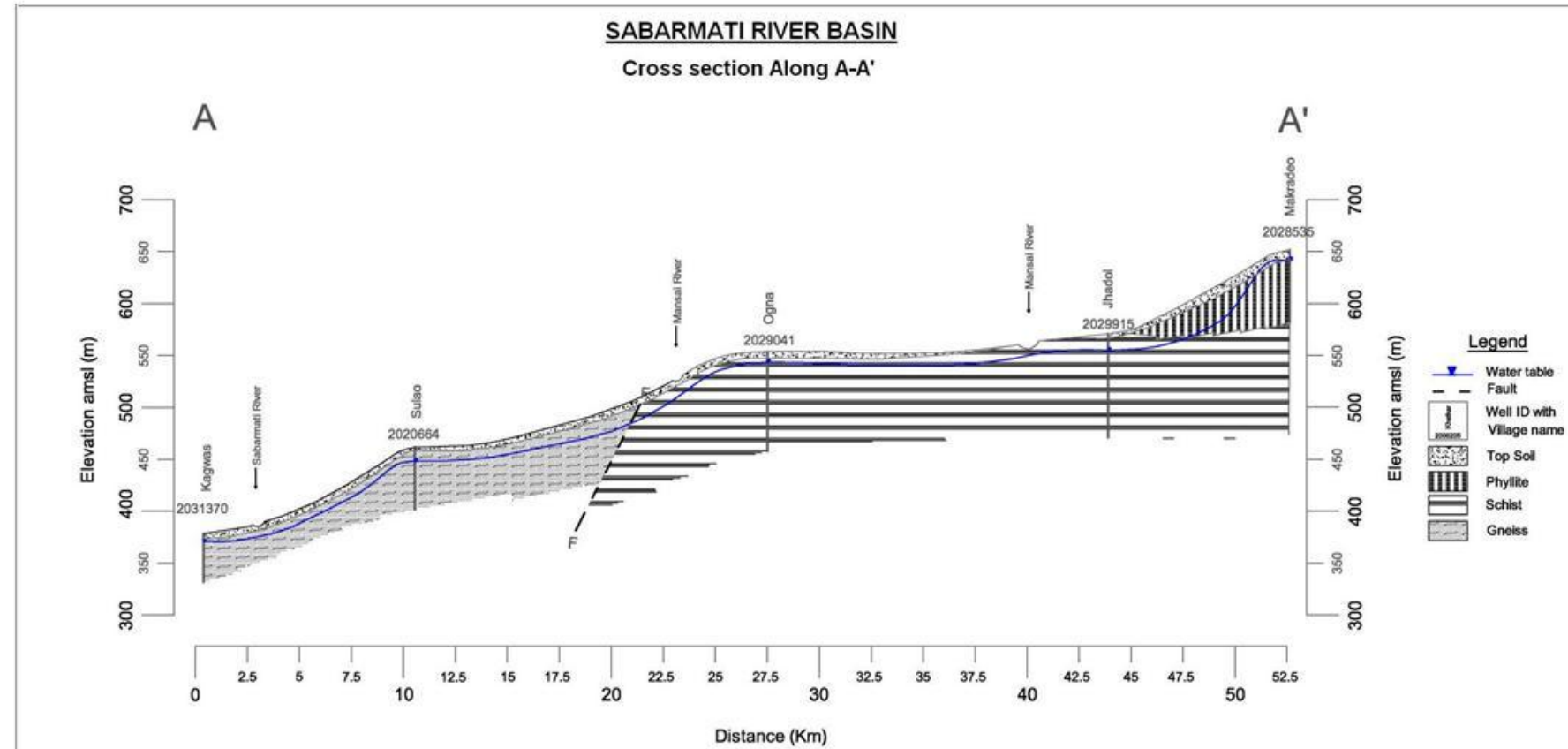
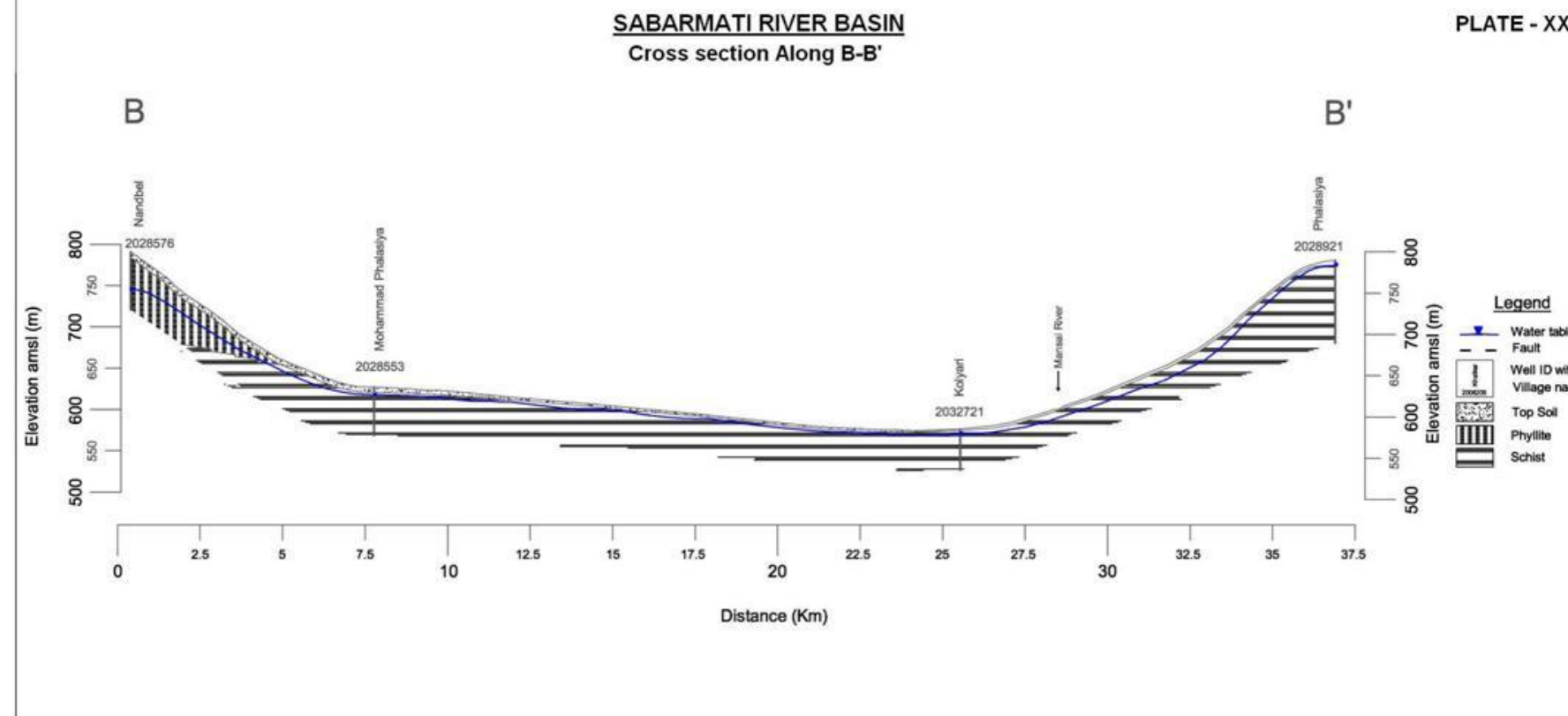


PLATE - XX



CROSS SECTIONS

SABARMATI RIVER BASIN

Section C-C':

The C-C' (Plate – XXI) section has been plotted across the southeastern isolated part of the basin trending NW-SE, covering a distance of about 42.5 km, broadly perpendicular to the Majhara and Vatrak Rivers. The lithologs of 8 boreholes along with surrounding well information has been considered while preparing the section. Except for a thin band of Quartzite in the northwestern part of the section, Phyllite is the only lithology present in the section and this is overlain by top soil throughout the section.

The water level in this section varies from 190 m amsl to 340 m amsl, parallel to surface topography.

Section D-D':

The D-D' (Plate – XXII) section is 22.5 km long trending in N-S direction, in the southern part of the basin. The lithologs of 5 boreholes along with surrounding well information is taken into consideration while preparing the section. The section reveals a monotonously single lithology i.e., Phyllite in all the wells that is overlain by top soil throughout the basin.

Ground water level varies from 210 m amsl to 300 m amsl, following the topography that slopes from north to south.

PLATE - XXI

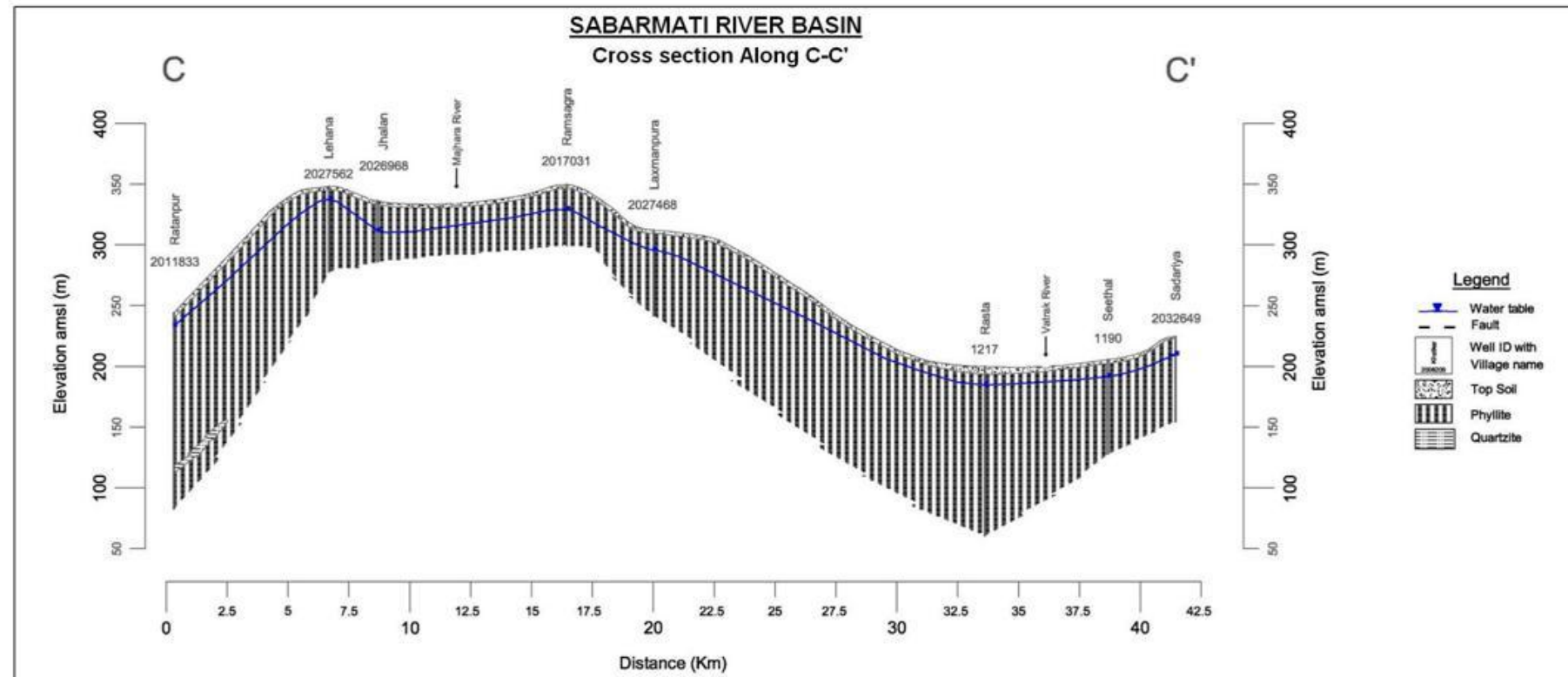
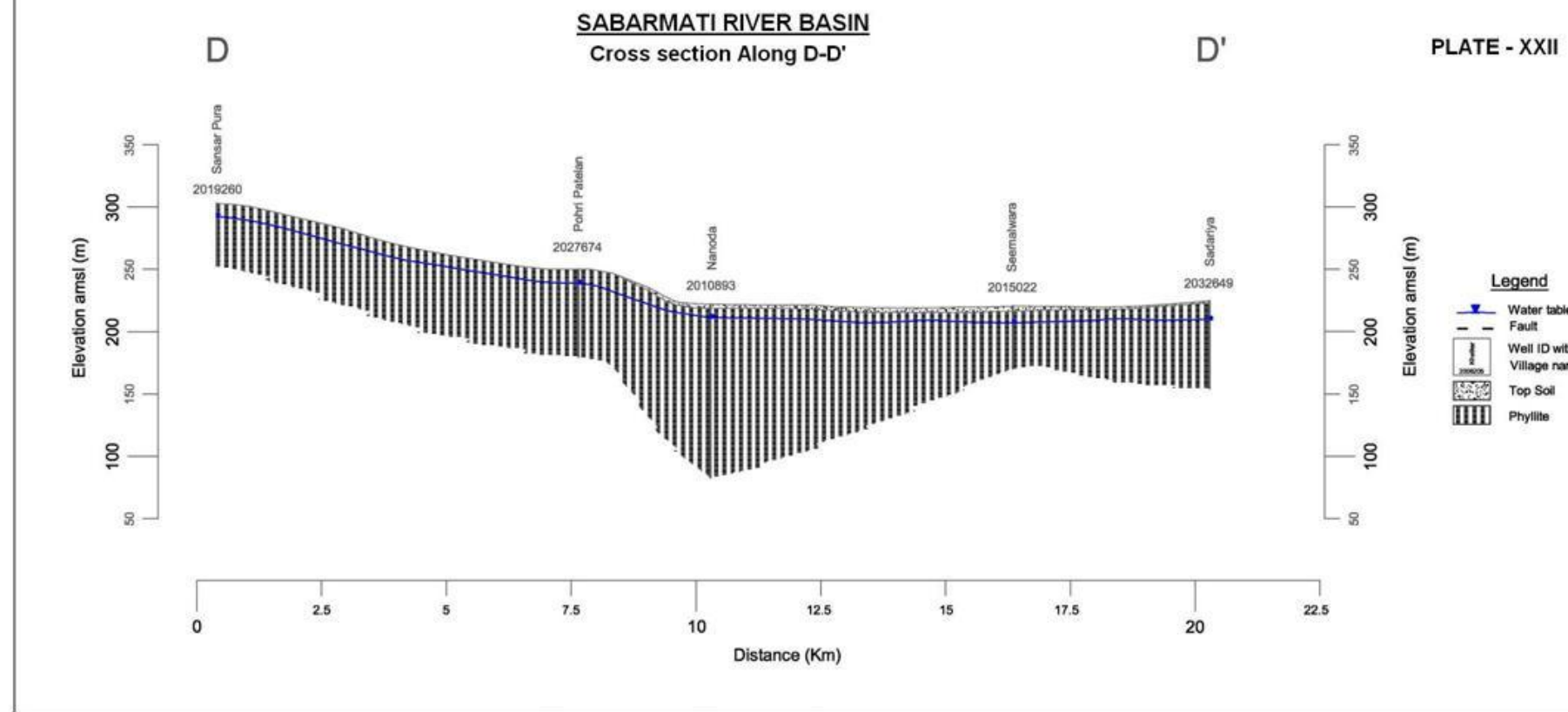


PLATE - XXII



3D MODEL OF AQUIFERS

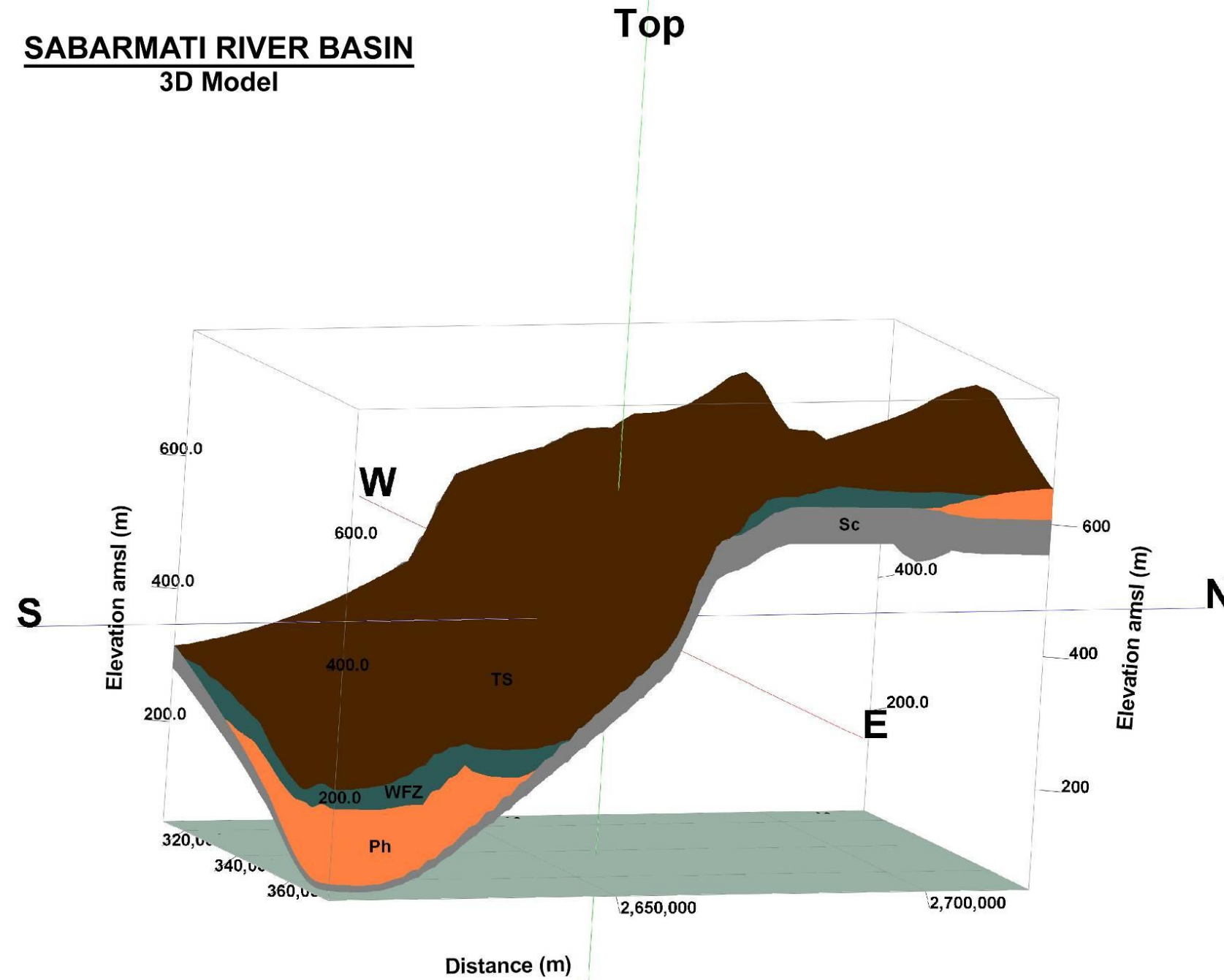
SABARMATI RIVER BASIN

The continuous litho-stratigraphic model has been developed for the Sabarmati River Basin using the data of scattered wells as input. 3D model depicts the sub-surface aquifer disposition of litho-stratigraphic units forming aquifers, aquicludes and aquitards in the area. Plate XXIII presents 3D model depicting the various litho-stratigraphic units in the entire river basin.

With this model it is apparent that beneath the top soil there is a persistent weathered and fractured zone of hard rock acting as unconfined aquifers, in south east and southeastern part of the basin. Predominantly, massive Phyllites and Schists are underlying the weathered and fractured zone in the area. Thickness of schist is high in north as compared to other area of the basin.

PLATE - XXIII

SABARMATI RIVER BASIN
3D Model



Legend

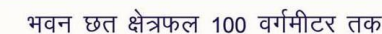
TS	Top Soil
WFZ	Weathered & Fractured Zone
Ph	Phyllite
Sc	Schist
G	Granite

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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